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AI regulation Made in Europe – a Public Value perspective

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Abstract

The exponential improvement of Artificial Intelligence (AI) technology and its adoption across business and personal applications impacts European Union (EU) fundamental rights and values. Recently, an increasing number of harmful incidents has sparked debates on soft and hard law approaches towards AI. Consequently, the EU has implemented an AI Act (AIA) as the first player globally. So far, research has focused on expert, private and public player preferences regarding AI ethical principles and regulation. Evidence on which values should guide the regulation of AI from a citizen perspective is missing despite value convergence being imperative to achieve acceptance of regulation and thus technology adoption. Therefore, this research paper focuses on EU citizens as the ultimate arbiter of value by applying Meynhardt's Public Value (PV) concept. Applying a sequential mixed methods approach we offer the first systematic analysis of values constituting the AIA and a comparison with empirically assessed value preferences of a German, Portuguese, and Finnish sample. Our findings are threefold. First, qualitative content analysis of the AIA revealed 33 value categories along four PV dimensions with a strong weight of *instrumental-utilitarian* (IU) followed-by *moral-ethical* (ME) values. Second, in a survey, which contained the derived AIA value inventory extended by Public Values and convergent ethical AI principles, citizens rated all values important and the ME dimension as significantly most important. Third, statistical testing on the value level proved varying preferences between countries and certain value pairs. The preferred values showed a more balanced distribution across the four PV dimensions than the AIA. As a result, the AIA meets citizen needs better than analyzed ethical AI guidelines, but the reverse focus of the IU compared to the ME dimension and a missing balance between the PV dimensions shows a gap between public preferences and the regulatory focus. Therefore, we suggest improving AI ethics and regulation by complementing it with PV theory and respective tools. In this way, relational dynamics between citizens and regulators could be better addressed leading to more effective, citizen-centered policies.

"The rise of powerful AI will either be the best or worst thing ever to happen to humanity."
(Stephen Hawking)

1 Introduction

The use, relevance, and impact of Artificial Intelligence in the realm of business has grown rapidly due to exponential technological improvements spanning applications like customer service, industrial manufacturing and critical fields like clinical diagnosis (Chazette & Schneider, 2020). With the introduction of the generative AI tool ChatGPT, AI now has also gained widespread public access. The technology is expected to contribute to a wide array of economic, societal, and environmental benefits by improving prediction, accelerating personalization, optimizing operations and resource allocation across the entire spectrum of industries and social activities (EC, 2021). At the same time, improperly designed and deployed AI may generate risks. Misinformation, propaganda, manipulation, job losses due to automation and security threats can cause material and immaterial harm to the public interest (EC, 2020). In Europe this directly impacts values the Union is founded on, and fundamental rights enshrined in the EU Charter of Fundamental Rights (Council of Europe, 2017). A rising number of incidents where AI harms society demonstrates this risk (O'Neil, 2017; Pasquale, 2020). Now, even liberal market poster children like Elon Musk lobbying for pausing the development and starting the regulation of AI systems (Future of Life Institute, 2023). Overall, the question is no longer if AI will change individual and collective experiences, values and thus society at large, but how we, as a society, want to live with it.

Naturally, this has sparked a heated debate among stakeholders which values AI development needs to follow and how AI needs to be controlled and/or regulated (Guerkaynak, Yilmaz & Haksever, 2016; Scherer, 2015). In consequence, many stakeholders involving experts, private organizations, and public institutions have published guidelines with value-oriented principles that are supposed to provide normative guidance for AI development, deployment, and use. However, the large number of guidelines complicates finding a common set of values that should govern AI (Rudschies et al., 2021), and the guidelines themselves are blamed to be ineffective (Hagendorff, 2020) and. The EU has answered the calls by spearheading the regulation of AI with the self-set goal to enhance the "individual and collective wellbeing" (HLEG, 2019, p. 11). Based-on Ethics Guidelines for Trustworthy AI (HLEG, 2019) the European Commission was quick to table a proposal for an AI Act (EC, 2021) recently approved by the European Parliament (EP, 2024). By this the EU is the first player worldwide to deviate from a soft towards a hard law approach (Abbott & Snidal, 2000).

Building on “intrinsically European and humanistic values” this has been promoted as a major step to ensure that “AI systems become human-centric, resting on a commitment to their use in the service of humanity and the common good, with the goal of improving human welfare and freedom” (HLEG, 2019, p. 4). To alter the AIA and ensure fertile grounds the EU has conducted a public consultation beforehand. However, only 17% of the participants were citizens, while 35% represented private organizations (EC, 2023). This indicates missing societal participation. Moreover, the EU’s AI policies have been criticized to have been watered-down by private companies (Benkler, 2019; Metzinger, 2019). In this case the relation between ethics and law would invoke a false sense of ethical responsibility for the public good (Anderson, 2022).

Research promotes participatory design to ensure AI’s contribution to the public interest (Delgado et al., 2023). However, technological and regulatory solutions are often implemented without critical input of society (Aizenberg & van den Hoven, 2020). Therefore, academia has called for an assessment whether current regulations are sufficiently grounded in societal values (Floridi et al., 2018). While several studies have analyzed preferences of state (e.g., Radu, 2021; Djeffal, Siewert & Wurster, 2022), non-state (e.g., Tallberg, Lundgren & Geith, 2024) and societal (e.g., Zhang, 2019) actors regarding hard vs. soft law approaches for AI research on which values policies should promote from a citizen perspective is missing. This is imperative success of policy measures is highly dependent on the extent value preferences of citizens are represented (König, Wurster & Siewert, 2023). Public denial of technology policies for genetically modified food, nuclear power, and vaccines stress this impressively (Zhang, 2022).

Taking an EU perspective, we answer the following research questions:

- Which values constitute the EU AI Act?
- What public values should AI regulation fulfill from the perspective of EU citizens?
- How do the EU AI Act values compare to value preferences of citizens and other stakeholders?

To answer these questions, we apply a mixed-methods approach (Creswell, 2003). In a sequential process values promoted and protected by the AIA are derived using qualitative content analysis (Myers, 2020; Mayring & Fenzl, 2019) and allocated towards the four public value dimensions of Meynhardt’s Public Value Theory (2015) as deductive themes. These values are extended by additional ones originating from the HLEG Ethics guidelines (2019) and Public Value Theory (Jorgenson & Bozeman, 2007; Meynhardt, 2009) and are subsequently assessed for their importance when regulating AI organizations in a quantitative survey conducted in Germany, Portugal and Finland.

Our results show 33 value categories that constitute the AIA with a strong weight of the instrumental-utilitarian dimension followed by the moral-ethical and the hedonistic-aesthetical and political-social dimension. All assessed value dimensions and value categories are deemed important for the regulation of AI organizations by society, yet the moral-ethical dimension is significantly more important than the other dimensions. Additionally, the instrumental-utilitarian dimension is more important than the political social dimension. Moreover, Stakeholder participation and Cultural heritage are significantly less important for the Finnish, German and Portuguese sample compared to the other tested values. This proves that the weight of the value-dimensions in the AIA does not fully match EU citizen preferences and that two important value categories (Citizen involvement; Compromise) are missing in the AIA. Last, citizens of all countries intent to use regulated AI with a higher likelihood than unregulated AI, where the Finnish sample has a higher likelihood to use unregulated AI.

The advantages of our research are threefold. First, our results provide regulators with an empirical analysis of the importance of certain values when regulating AI organizations. In addition, we have found country differences that can improve communication effectiveness when promoting the AIA. Taking these preferences into account could strengthen the grounding of regulation in societal values which is expected to increase acceptance. Likewise, organizations can factor-in value-preferences of their end-users when developing, deploying, and using AI to increase adoption rates of their products and services. Second, we introduce Public Value Theory as a valuable complement to overcome the shortcoming and criticism of AI ethics. The Public Value Scorecard is introduced as a tool to measure the public value impact of EU AI policies with future research recommended. Third, this paper contributes to multiple fields of research. It adds to Public Value Theory by applying a mixed-methods approach including evaluations by the public and researching the theories' applicability in the legal realm for the first time. Thereby, we counteract the criticism of diverse empirical research scarcity in the public value domain (Hartley et al., 2017, Meynhardt & Bartholomes 2011; Williams and Shearer 2011). Moreover, it provides new evidence on actor preferences towards AI regulation in the growing field AI governance research, which so far has focused mainly on states and institutions (Tallberg, Lundgren & Geith, 2024). Last, we test the application of content analysis to legal texts, which has been promoted to make legal analysis more consistent with epistemological underpinnings of social science research (Hall & Wright, 2008).

Before beginning with our analysis, it is important to establish a definition of AI. Currently, there is no universally accepted definition especially between different fields or disciplines (Larsson, 2021). Rather than referring to concrete applications, it reflects recent technological developments that encompass a variety of technologies. As our paper takes an EU perspective we leverage the definition of the AIA, which to our knowledge is the most recent definition by the European Commission (2024, p. 97):

"An AI system is a machine-based system designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments."

It is important to view AI in its applied context and thus in interaction with societal structures, values, norms, and rights. Mostly, because, as stated by the Commission, all AI systems are dependent on human generated input data.

2 Artificial Intelligence for the common good

Many AI stakeholders like the EU were quick to postulate the key task of society is to ensure that the technology serves the common good (HLEG, 2019). Norbert Wiener (1960), the inventor of cybernetics, used the analogy of Goethe's Sorcerer's Apprentice to remind people of the importance that the purpose we program into a machine matches the one we really desire. Regarding AI, this means that goals and behaviors of highly autonomous systems should be aligned with human values throughout their operation (Asilomar, 2017; Russell, 2019). This is especially true as AI systems develop at a pace that "increasingly prohibits humans from evaluating whether each action is performed in a responsible or ethical manner" (Allen et al., 2005, p. 149). This endeavor involves three challenges with values at the core: First, we need to agree on what an AI system is morally ought to do. Second, these values need to be formally encoded into artificial agents to reliably do what they are supposed to do (Gabriel, 2020). Third, legislators need to find effective policies to ensure that AI practitioners act according to those normatively aligned and technical feasible values. Existing literature on regulatory preferences shows, that regulatory effectiveness via the acceptance and adoption of policies is determined by the extent to which policies align with citizens' preferences (e.g., Ingold, Stadelmann-Steffen, & Kammermann, 2019; Wüstenhagen, Wolsink, & Bürer, 2007).

What sounds straightforward in theory raises two critical challenges in practice: Which values should AI be aligned with? Who decides on these values? In this chapter we answer these questions by summarizing the current state of the literature on AI normative value alignment. Public Value theory is introduced to provide solutions for identified shortcomings of AI ethics. Last, we give an overview on the EU's regulatory framework towards AI.

2.1 AI ethics

Two factors limit finding a true set of morally right principles that should govern AI: The absence of a single moral theory which is true and the fact that we live in a pluralistic world in which humans hold a variety of reasonable but partially contrasting beliefs about value (Rawls, 1999). Political theory offers a solution based-on the conviction that people are free and equal and can have different values and opinions yet are still able to agree on principles that should govern a specific subject matter or set of relationships (Gabriel, 2020). The agreement takes the form of overlapping consensus on a set of principles irrespective of varying beliefs about the nature of morality (Rawls, 2001). Regarding AI, the most prominent approach to find overlapping consensus on existing values that should provide guidance to the development, use and marketing of AI are convergence studies on AI ethics.

Over the last year AI ethics has experienced a true renaissance with virtually all actors with a connection to technology policy authoring or promoting AI ethics principles (Fjeld et al., 2020; Schiff et al., 2021). Alone since 2017 over 60 public players published more than 700 documents to set out their visions on ethical AI (Foffano et al., 2023). Ethics guidelines share a basic purpose but vary in their targeted audiences both in terms of geography and player types, scope and depth. Synthesis of the included ethical principles to find common ground therefore remains limited. Nevertheless, the few existing studies have determined a normative core of a values-based approach towards AI ethics and governance with an increasing convergence in promoted values-based principles over time (Fjeld et al., 2020). The most cited studies have been conducted by Floridi et al. (2018), Jobin et al. (2019), Fjeld et al. (2020) and Hagendorff (2020). Rudschies et al. (2021) have compared these existing meta-analyses and extended them by a content analysis of 40 documents. The most up-to-date analysis is from Correa et al. (2023).

In their theoretical work Floridi et al. (2018) lay-out policies, best practices and recommendations for ethical AI. Besides, they analyzed 20 documents from multiple stakeholders with a focus on public organizations that have stated 47 principles to ground their recommendations in research. The authors find coherence on four core principles commonly used in bioethics, namely Beneficence, Non-maleficence, Autonomy, and Justice. Bioethics resembles digital ethics in its quest to ecologically deal with new forms of agents, patients, and environments. Another principle added by the authors is Explicability, which combines Intelligibility and Accountability. Jobin et al. (2019) have analyzed a multi-stakeholder sample from around the world except for Latin America that contained 84 sources. They identified global convergence around the principles of Transparency, Justice and fairness, Non-maleficence, Responsibility and Privacy. In a similar quest, Fjeld et al. (2020) conducted a meta-analysis of 35 civil society and multistakeholder documents from all continents. Despite the large set of documents, a solidifying normative core of eight key themes (Privacy; Accountability; Safety and security; Transparency and explainability; Fairness and non-discrimination; Human control of technology; Professional responsibility; Promotion of human values) of a principle-based approach to AI ethics and governance was found. Hagendorff (2020) conducted his semi-systematic evaluation of AI ethics guidelines with an equally small sample as Floridi et al. (2018) containing 22 documents not older than five years with a pure focus on AI by excluding robotics and data science related documents. He clusters actors into public (government), private (industry; non-profit) and expert (science) players. Out of 22 identified principle-sets, six were stated the most: Privacy; Fairness, non-discrimination, justice; Accountability; Transparency, openness; Safety, cybersecurity; Common good, sustainability, well-being. Last, Correa et al. (2023) conducted the most recent analysis with the largest sample of 200 governance policies and ethical guidelines for AI usage published by public bodies, academic institutions, private companies, and civil society organizations worldwide. Out of 17 resonating sets of principles prevalent in these documents the top five are Transparency, explainability, auditability; Reliability, safety, security; Justice, equity, fairness, non-discrimination; Privacy; Accountability, liability.

	Floridi et al. (2018)	Jobin et al. (2019)	Fjeld et al., (2020)	Hagendorff (2020)	Rudschies et al. (2021)	Correa et al. (2023)
Included		Transparency (87%)	Transparency & Explainability	Transaprency & Openness (73%)	Transparency (98%)	Transparency, Explainability, Auditability (83%)
	Explicability			Explainability, interpretability (45%)	Explainability (78%)	
	Justice	Justice & Fairness (81%)	Fairness & non-discrimination	Fairness, Non-dicrimination, Justice (82%)	Non-discrimination & fairness (83%)	Justice, Fairness, Equity, Non-discrimination (76%)
	Non-maleficence	Non-maleficence (71%)			Non-maleficence (68%)	
		Responsibility (71%)	Professional Responsibility		Responsibility (78%)	
		Privacy (56%)	Privacy	Privacy (82%)	Privacy (88%)	Privacy (69%)
	Beneficence	Beneficence (49%)		Common good, sustainability (73%)	Beneficence (68%)	Beneficence, non-maleficence (47%)
			Accountability	Accountability (77%)	Accountability (75%)	Accountability, Liability (67%)
			Safety & Security	Safety & Cybersecurity (73%)	Safety & Security (75%)	Reliability, Safety, Security (78%)
			Human control	Human oversight, auditing (55%)	Human control & determinism (70%)	
Excluded	Autonomy	Freedom & Autonomy (40%)			Autonomy (35%)	Freedom, Autonomy, Democracy, Sovereignity (54%)
			Promotion of human values		Promotion of human values (63%)	

Table 1, AI ethical principles convergence across meta-analyses; Source: Own representation

To find the most prevalent principles stated in the above-mentioned studies, we have excluded all principles that were not mentioned in at least three of the listed meta-analyses and <50% of the documents within the respective meta-analysis in case this information was available. As table 1 shows we find ten principles that are most often stated to promote ethical AI: Transparency; Explainability; Justice, fairness, non-discrimination; Non-maleficence; Responsibility; Privacy (protection); Beneficence, well-being; Accountability, Liability; Safety, Security; Human control/agency/oversight. Our findings replicate Rudschies et al. (2021) as only the Justice, fairness, non-discrimination principle is mentioned in all analyses. Furthermore, we find that Transparency; Explainability; Privacy; Beneficence, well-being are stated in five out of six documents. Together these principles seem to represent the overlapping consensus of ethical AI values.

Despite some convergence in the stated papers, Rudschies and colleagues (2021) underline that the meta-analyses results show significant differences regarding convergent principles. Next to variation in datasets and applied methods it has been hypothesized that differing stakeholder priorities fuel this diversity. Most documents originate from public and private institutions, followed by CSOs/NGOs, non-profit organizations, and academia (Fjeld et al., 2020; Jobin et al., 2019; Correa et al., 2023). Regarding preferences for principles there are significant differences between public and expert compared to private ethics guidelines. Rudschies et al. (2021) find that public and expert actors place more emphasis on fundamental rights and democratic principles as well as the principles of Explainability, Accuracy, completeness, and Sustainability than private players. These principles have been proposed by at least twenty percentage points fewer guidelines from private actors than by the other actor types. Private actor documents show fewer principles, with a focus on those that align more closely with existing technical solutions or legislation. Comparing public to expert actors also reveals some minor differences. Public actors emphasize Non-discrimination and fairness, Freedom, Democracy and the rule of law, while experts move the individual into the center of attention with principles like Responsibility, Accountability, Non-maleficence, Contestability and Privacy. Overall, the authors find that all player types favor soft-law approaches. 56% of analyzed documents contain non-binding recommendations to (multiple) AI stakeholders, 24% possess voluntary self-regulatory/self-commitment style guidelines, while only 20% propagate a form of governmental regulation. A recent study in the EU by Tallberg and colleagues (2024) who have analyzed data collected by the EC for a public consultation on the White Paper on AI (EC, 2023) shows that in the EU state and non-state actors are in favor of regulating AI’s development and use. However, private actors tend to favor a laxer regulation.

Overall, AI ethics guidelines are criticized for several reasons. First, most guidelines originate from governmental institutions and private corporations, while the public is underrepresented. A true consensus in values that serve to guide AI ethics needs to represent the preferences of all AI stakeholders and especially the public (Gabriel, 2020). In addition, studies found high divergence in how these principles are understood, why they are deemed important and how they are promoted to be implemented (Jobin et al., 2019). On cause is that promoted principles and values represent abstract and vague concepts leading to different interpretations (Correa et al., 2023; Fjeld et al., 2020; Jobin et al., 2019). Therefore, guidelines are blamed to act as placeholders concealing potential normative and political disagreement and thus not reflecting meaningful consensus (Mittelstadt, 2019). Moreover, different streams of research have questioned the effectiveness of AI ethics to safeguard the potential negative influences of the technology on society (Hagendorff, 2020; Züger & Asghari, 2023, McNamara et al., 2018). Procedural weakness results from the tendency of guidelines to focus on aspirational needs at a general level (Larsson, 2021) and missing integration into larger governance ecosystems (Jobin et al., 2019; Mittelstadt, 2019). This allows private players to exploit ethics guidelines to discourage truly binding legal frameworks (Hagendorff, 2020). Faced-by the call of the public for responsibility and regulation, private technology organizations therefore try to influence the regulation process. They do so by incorporating ethical considerations into public relations work, or by adopting ethical self-commitments. Industry- or science- led guidelines serve to suggest to legislators that no specific laws are necessary to mitigate AI risks (Calo, 2017) and that accountability can be shifted from institutions legitimated by the public to a self-governed science or industry (Hagendorff, 2020). Furthermore, they are utilized to delay regulation and focus public debate on abstract problems and technical solutions (Greene, Hoffmann & Stark, 2019). This is especially worrying as AI can be classified as a technology where industry races ahead of academia and governments as the majority of innovation is achieved by private institutions (Maslej et al., 2023). Last, some values like gender diversity, labor rights and potential negative impacts like misinformation and AI militarization are missing in the ethics discourse (Hagendorff, 2020).

Despite the criticism it would be too early to declare a consensus in opinions about ethical AI values. Yet, to improve and become accepted by society, value-based principles need to offer concrete guidance, represent preferences of people affected by the technology and be stable over time (Rawls, 1987).

2.2 Public Value Theory

Public Value Theory (PVT) offers solutions to overcome the shortcoming of AI ethics that risks overlooking humans as the ultimate arbiter of value. Initially coined by Moore (1995) Public Value refers to the contribution of an organization to society and has gained popularity as a guiding concept in theory and practice (Hartley et al. 2017; Williams & Shearer, 2011). Since then, four major public value approaches and formal theories have evolved by extending the focus of public value creation. The first stream of public value studies directly builds on Moore (1995) focusing on how public managers can better serve public value as a counterpoint to value generated by private businesses. A second stream of studies (e.g., Jørgensen & Bozeman, 2007; Bozeman, 2007; Nabatchi, 2012) extended the concept beyond public managers by assigning responsibility for society to both private and public managers. In a positivist and normative approach, the main research objective lies in identifying, defining, and enacting values that qualify as public values based-on societal consensus (Meynhardt & Jasinenko, 2020). In their first attempt to provide the concept of public value with an empirical basis, Jørgensen and Bozeman (2007) created 20 value sets from 72 values and assigned them to seven affected aspects of public administration or organization. Furthermore, they identified eight nodal values which have many related values and thus occupy a central position in a value network. Third, Benington (2011) draws attention to the public sphere or realm by theorizing that only what is valuable to society, revealed through an effective democratic process, can contribute to the public, and thus creates value. Fourth, Meynhardt (2009) has developed a psychological public value approach based-on need fulfillment that shares key assumptions with the previous research streams. He reconsiders how and for whom value is created by focusing on individuals enacting value while they perceive and evaluate, interpret and reason to construct their psychological reality (Meynhardt & Fröhlich, 2019). Thereby he enriches macro-foundations of value relating to collective properties by stressing that true value creation involves direct human appreciation. Overall, all research streams have deviated from the perception that public value is simply produced and only perceived by society to an understanding of public value being co-created within its ecosystem (Meynhardt & Jasinenko, 2020). This paper focuses on Meynhardt's (2009, 2015) public value conceptualization which is defined in detail below.

Meynhardt (2009, p. 205) defines public value as "anything people put value to with regard to the public". Moreover, public value creation is "…situated in relationships between the individual and society, founded in individuals, constituted by subjective evaluations against basic needs, activated by and realized in emotional-motivational states, and produced and reproduced in experience-intense practices" (Meynhardt, 2009, p. 212). As a result, three elements are key to the understanding of his theory, which are described in detail in the following: i) individual subjective evaluation of value, ii) the dynamic interplay between individuals and the public, iii) embedment of value evaluation in basic, psychological needs.

Referring to Heyde (1926) Meynhardt's theory regards value as the result of a relationship between a subject that is valuing an object and the valued object. Hence, public value is a subjective evaluation of a public made-up by individuals bound to each individuals' subjective relation to the object of evaluation and thereby neither purely subjective nor objective. In other words, public value cannot just be delivered by organizations or institutions but needs to be perceived and valued by individuals considering the public (Meynhardt & Jasinenko, 2020). Thus, Meynhardt (2015, p. 149) states that "[p]ublic value as a collectively shared value is not constructed as a sum of individual values, but their common and overlapping meaning about the quality of a relationship involving the public". As an operational fiction public value emerges from interactions at the micro-level, stabilizes and subsequently serves as a synergetic order parameter serving as a point of reference for social evaluations (Meynhardt et al., 2016). This evaluation is embedded in basic needs that concern deficits – felt discrepancies between an actual state and a desired state that results in a human motivation to act (Meynhardt, 2009). Value is created for a subject when basic needs are fulfilled by the object evaluated and destroyed if not (Meynhardt & Fröhlich, 2019). According to Epstein (2003) cognitive-experiential self-theory, four equally important, non-normative basic needs exist: (1) positive self-evaluation, (2) maximizing pleasure and avoiding pain, (3) gaining control and coherence over one's conceptional system and (4) positive relationships (Epstein, 1993; Gomez & Meynhardt, 2014). Despite varying importance over time these basic needs apply to all people regardless of their age, gender, or cultural background and build the explicit or implicit foundation for human motivations and evaluations (Epstein, 2003). Meynhardt translated each basic need into a basic value dimension, which is ultimately the basis for the evaluation in Public Value Theory. Firstly, the moral-ethical dimension, which is based on the human need for positive self-evaluation. The moral-ethical dimension represents "a collectively shared value ascribed to personhood and what it means to be human" (Meynhardt, 2015, p. 154). Second, the hedonistic-aesthetic dimension represents "a collectively shared value ascribed to pleasure and what it means to create a positive experience" (Meynhardt, 2015, p. 154). The hedonistic-aesthetic dimension builds on the basic human needs of maximizing pleasure and avoiding pain (Meynhardt, 2009). Third, the utilitarian-instrumental dimension which builds on the need for gaining control and coherence over one's conceptional system and represents "a collectively shared value ascribed to utility and what it means to create a benefit efficiently."

(Meynhardt, 2015, p. 154). The fourth and last dimension, is the political-social dimension, which is conceptualized from the need for positive relationships and represents “a collectively shared value attached to social relationships and what it means to establish positive group relations” (Meynhardt, 2015, p. 154).

PVT offers three advantages to guide the development, deployment, and use of AI. First, it shifts the focus of value creation back to the individual. Many major academic, governmental and industrial organizations have included participatory governance as a key success factor of their AI development frameworks (OECD, 2019; IEEE, 2019, EC, 2018). However, AI’s technical complexity and speed of innovation pose challenges to public participation. Discourse about AI governance has thus been criticized for being opaque, expert-led and vulnerable to be influenced by powerful entities (Ulnicane et al., 2021). Experts can help to guide value judgements, yet ultimately value must be perceived and experienced among the publics rather than delivered. Policymakers should therefore try to better understand underlying values that shape public attitudes towards AI and its regulation. Public value as a necessary fiction how the quality of social relations are or could possibly become, can be leveraged to gain detailed understanding about public opinions (Meynhardt, 2021).

Second, PVT states that the public is not just a simple addition of individual values and self-interests but a macro-level phenomenon connecting the individual to a social entity like community or society (Meynhardt & Fröhlich, 2019). In contrast to “aggregationist”, bottom-up approaches of social choice theory promoted for AI values (Prasad, 2018) it considers the full scope of what people put value into. It thus adds nuance when thinking about how to evaluate aggregate claims. Instead of the need to rely on and choose between different interests of individuals or experts, PVT allows to guide AI regulation based-on evaluative judgments against universal needs (Moor, 1999). Positive evaluations are expected to increase societal acceptance. On the other hand, societal resistance and counteraction might form in the long run as a result of unmet basic human needs. Especially regarding innovation this has happened often in the past. Examples include genetically modified food and nuclear power (Siegrist, 2000).

Third, as an overall consequence, decision-makers including regulators and organizations of AI technology need measurement tools to make valid predictions on the value of their choices perceived by different public(s), as well as on areas of improvement (Meynhardt et al., 2017). Public value is a matter of attitude and can be measured and analyzed by asking people what they think and feel as well as observing what they do. Meynhardt's Public Value Scorecard framework (2009) can be leveraged to measure how individual and societal well-being is affected by the regulation of organizations providing AI products and services.

2.3 The EU AI approach

In 2018 the EU has formulated its vision of an ethical, cutting-edge AI Made in Europe. Three key aspects underpin the vision. First, increasing public and private investments in AI. Second, the promotion and preparation for the digital transformation of the EU's economy and society. Third, an ethical and legal framework to strengthen European values in AI development, deployment, and use. In this regard, which is the focus of this paper, policymakers pledged to develop recommendations and legislation to ensure that Europeans can benefit from modern technologies developed and functioning according to EU values and principles.

To support the vision's implementation 52 experts forming a High-Level Expert Group on AI (HLEG) published non-binding Ethics Guidelines for Trustworthy AI (2019). The guidelines consist of three layers and their corresponding features. First, three components that need to be met throughout the entire lifecycle of AI to be perceived as trustworthy. AI should be (1) lawful, complying with all applicable laws and regulations, (2) ethical, ensuring adherence to ethical principles and values, and (3) robust, both from a technical and social perspective. Second, principles of (1) Respect for Human Autonomy, (2) Prevention of Harm, (3) Fairness and (4) Explicability. The principles have been revised based-on public feedback and are grounded in the EU Charter of Fundamental Rights (2000) and its common values that constitute these rights: dignity, freedom, equality and solidarity as well as citizen's rights and justice. Third, seven non-hierarchical requirements that must be achieved with the help of technical and non-technical methods in the development, deployment, and use of AI systems to realize trustworthy AI. These requirements are (1) Human Agency and Oversight, (2) Technical Robustness & Safety, (3) Privacy & Data Governance (4) Transparency, (5) Diversity, (6) Societal & Environmental Well-being, and (7) Accountability.

The HLEG states that that all requirements are of equal importance and acknowledges that as a result conflicts and (political) tensions between principles and requirements can arise. To solve ethical dilemmas and trade-offs AI practitioners should apply a reasoned, evidence-based manner rather than applying intuition or random discretion. Furthermore, end-users and society should be informed about the principles, strive for alignment with societal values and control for their adherence by deployers. In line with the EU fundamental commitment to democratic engagement, due process and open political participation, the experts propose to establish methods of accountable deliberation to deal with such tensions (HLEG, 2019). As with Ethics Guidelines in general, several topic experts including members of the HLEG themselves raised concerns on the true value of such regulation for society. Reasons focus on the composition of the HLEG as only two of 52 members represented civil society. Thus, the document is judged to “(…) further sideline and marginalize community and domain voices, and (…) to reify an elite club of AI and society experts to the detriment of those with connection to harms and issues that technologies exacerbate” (Veale, 2019 p. 8).

Since then, the EU has deviated from its soft-law approach. On the 13th of March 2024, the European Parliament has approved the AIA (EC, 2021, 2024) as the first comprehensive legal framework on AI worldwide. The Act is promised to improve the functioning of the internal market, ensure adherence to Union values, and protect citizens against adverse effect of AI, while stimulating innovation. It is therefore promoted to become a unique selling point for EU member countries, their companies, and citizens. While several authors have celebrated the pioneering efforts of the EU (e.g., Battista, 2021) others criticized missing ethical grounding despite the claim that the regulations ensure the protection of the HLEG’s ethical principles (Anderson, 2022) and lacking public participation during the development of the Act. Surveys showing that ~70% of interviewed European citizens are not aware of the, by then proposed, AIA provide evidence for the critique (Foffano et al., 2023). This is despite the EU’s goal to drive its social agenda as part of its trade policy as “forging collective preferences’” (EP, 2007). Extensive research has demonstrated the importance of public involvement to establish legitimacy of value-laden expert assessments (Mitchell et al., 2006).

This raises three research questions:

- Which values constitute the EU AI Act?
- What public values should AI regulation fulfill from the perspective of EU citizens?
- How do the EU AI Act values compare to value preferences of citizens and other stakeholders?

3 Research Methodology

To answer these research questions, a sequential mixed methods approach combining qualitative and quantitative approaches was used (Creswell, 2003). The research flow is depicted in figure 1. We combine qualitative, inductive data collection through a literature review and qualitative content analysis (Myers, 2020; Mayring and Fenzl, 2019) with a quantitative assessment via survey. This design allows to explore which values constitute the AIA as well as assessing their importance by asking the public about their perceived relevance (Creswell, 2003).

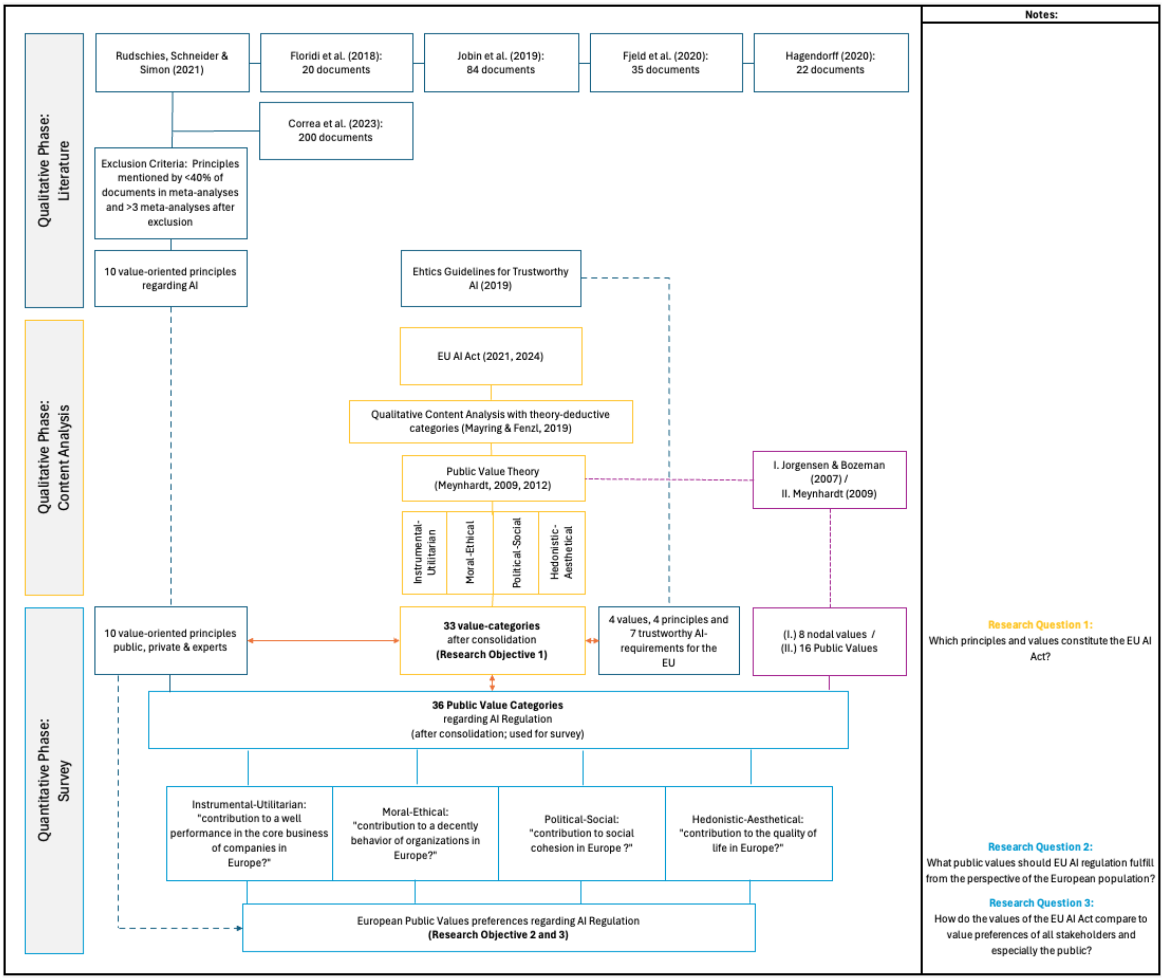


Figure 1; Research Flow; Source: Own representation

Values constituting the AIA (2021, 2024) were analyzed by applying qualitative content analysis (Mayring & Fenzl, 2019). Originating from social sciences this is a traditional approach to textual analysis (Berelsen, 1952), yet relatively new to the analysis of legal documents. Nevertheless, it is promoted to make legal analysis more consistent with epistemological underpinnings of other social science research (Hall & Wright, 2008). Advocates even claim that the method points toward the holy grail of true legal science (Reed, 1967). Research analyzing if content of law is adequate or suitable for the task and interests involved has found to be valuable (Macaulay, 1963; Ellickson, 1991; Kurkchiyan, 2009 in Galligan, 2010) and effective to test whether issues with proposed laws exist (Webley, 2010). In addition, the approach has been recommended to understand values, their meaning and relations conveyed in texts (Waerras, 2022). Structural qualitative content analysis seeks to analyze the meaning of written documents by systematically allocating the content to pre-determined, detailed categories and then interpreting the outcome (Payne & Payne, 2004). Descriptive, semantic coding was conducted independently by two researchers. We followed a two-step process combining deductive and inductive approaches. First, deductive categories, which are developed in advance, are applied to parts of the analyzed text as *themes* for which categories are developed deductively and inductively throughout the text-evaluation (Mayring and Fenzl, 2019). The qualitative content analysis was carried out according to the following framework conditions: individual words served as the coding unit, the context and evaluation unit were the paragraphs of the AIA. The overall process included a pilot phase, an initial evaluation phase to create a longlist and a second evaluation phase. For our research purpose we used the four basic value dimensions of Meynhardt (2009, 2015) and Meynhardt and Jasinenko (2020) as a theory-deductive theme system with a nominal scale level. Furthermore, we leveraged the ten most frequently stated value-oriented principles of AI ethics guidelines[1], four ethical AI principles, six common values and seven trustworthy AI requirements promoted by the HLEG (2019), eight nodal public values of Jorgensen & Bozeman (2007)[2] as well as adjusted 16 public values by Meynhardt (2009)[3] as deductive value themes. By clustering and eliminating identical values, we reduced the number of public values across the sources to 27. Considering these public values and the definitions of the selection criteria as words and the level of abstraction of the of the category system and the units of analysis each on the paragraph-level, formation of the inductive categories from the material was started. In the first round of coding, inductive categories including their count were derived from the text. By revisiting the 152 generated codes a hierarchical system was developed extending the public values by nine categories. The resulting 36 public values represent the basis of values which is tested for generalizability in the subsequent survey. Second, it was checked whether the codes can be summarized and reduced. All codes applied less than five times were excluded leading to the exclusion of three value categories that originated from the Public Value literature. This led to final 33 value categories built from 99 codes which are shown in detail in the appendix. Again, code counts were noted to enable a quantitative assessment.

As stated, the derived 36 public values were tested for their relevance in the view of European citizens as value convergence informs the acceptance of regulation by society. When the variables for quantitative testing are unknown, exploratory sequential mixed methods are recommended (Creswell & Clark, 2011). The applied approach allowed to use the value categories of the qualitative content analysis as input factors for a quantitative study aiming at deductive results (Morgan, 2007). We followed a purposive sampling approach on a country-level, which is the most common form of non-probabilistic sampling (Guest et al., 2006). The

[1] *Responsibility* assigned to *Justice*
[2] *Openness-Secrecy* changed to *Data privacy*
[3] *Service quality* changed to *User orientation; Openness-Secrecy* changed to *Data privacy; Social innovation* changed to *Innovation; Reliability* assigned to *Robustness*; Exclusion of *Beauty of public spaces* & *Self-initiative*

sample was recruited via the platform pollfish (www.pollfish.com) which relies on Random Device Engagement (Rotschild & Konitzer, 2020). With proper statistical adjustment non-probability sampling via pollfish is a valuable tool in various research domains also compared to probability-based studies (Goel, Obeng & Rothschild, 2015). Our final sample was aged between 18 and 73 and included participants from Portugal (n = 80), Germany (n = 76) and Finland (n = 55) after excluding participants with a response time <5min, with a wrong answer in the control question and/or no variation in responses for value importance from the overall sample of n=258. The countries were chosen as a public consultation conducted by the European Commission (EC, 2023) revealed that actors based in Finland held considerably more AI-friendly views and were less in favor of regulation than others. The opposite was found for actors based-in countries like Portugal, while Germany showed moderate values (Tallberg, Lundgren & Geith, 2024). 67% of the participants were male and 37% female. Most of the sample held a higher education degree (27% Bachelor, 34% Master and above) while 21% had either a completed professional training or technical baccalaureate and 13% held a high-school diploma. Regarding jobs 52% perform mainly cognitive or analytical tasks, 24% mainly manual or physical tasks and 24% mainly social or people-oriented tasks. Participants reported that they have a “basic idea of what AI is” (42%, n=87) or “good understanding of how AI works and where it is used” (35%, n=73). Regarding the frequency of using AI most respondents stated dealing with AI almost every workday (27%, n=56), once a week (26%, n=53), less than once a week (19%, n=39), or on a daily basis (15%, n=30).

Answers were collected between April 28, 2024, and May 08, 2024. The online questionnaire started with standard sociodemographic questions (e.g., gender, age, ethnicity political orientation). In addition, several questions were posed to gain insights on attitudes towards, knowledge about and engagement frequency with AI after introducing its definition analogous to the one outlined in chapter two. AI expertise and engagement frequency were assessed applying the same scales as Laupichler et al. (2022). Next, participants were asked to judge the importance of personal for the regulation of AI organizations on a six-step Likert-scale from “Very unimportant” (coded as 6) to “Very important” (coded as 1) (Saunders et al., 2009). Last, general attitudes towards AI and AI regulation (Poortinga & Pidgeon, 2005; Liu, Yang & Xu, 2018) and intention to use, purchase and recommend regulated as well as unregulated AI systems in the future was assessed (Choi & Ji, 2015; Liu, Yang & Xu, 2018).

4 Results

To answer which principles and values constitute the AIA a qualitative content analysis (Mayring & Fenzl, 2019) was conducted. As a result, a complete, hierarchical category system of the AIA (EC, 2021, 2024) for the four public value dimensions and twelve sub-dimensions as theory-derived themes including the associated frequencies of codes was developed. As a holistic concept the public value dimensions (Meynhardt & Jasinenko, 2020) encompass all basic human needs and thus provide a comprehensive framework for capturing the values of the AIA. The following figure 2 shows all 33 value categories of the AIA and their respective count in the document, which were coded into the four public value dimensions and twelve sub-dimensions. A detailed table including exemplary quotations can be found in the appendix. The 33 value categories were based-on sub-codes and assigned to the four public value dimensions as well as twelve value-items based on Meynhardt and Jasinenko (2020) as shown in figure 2.

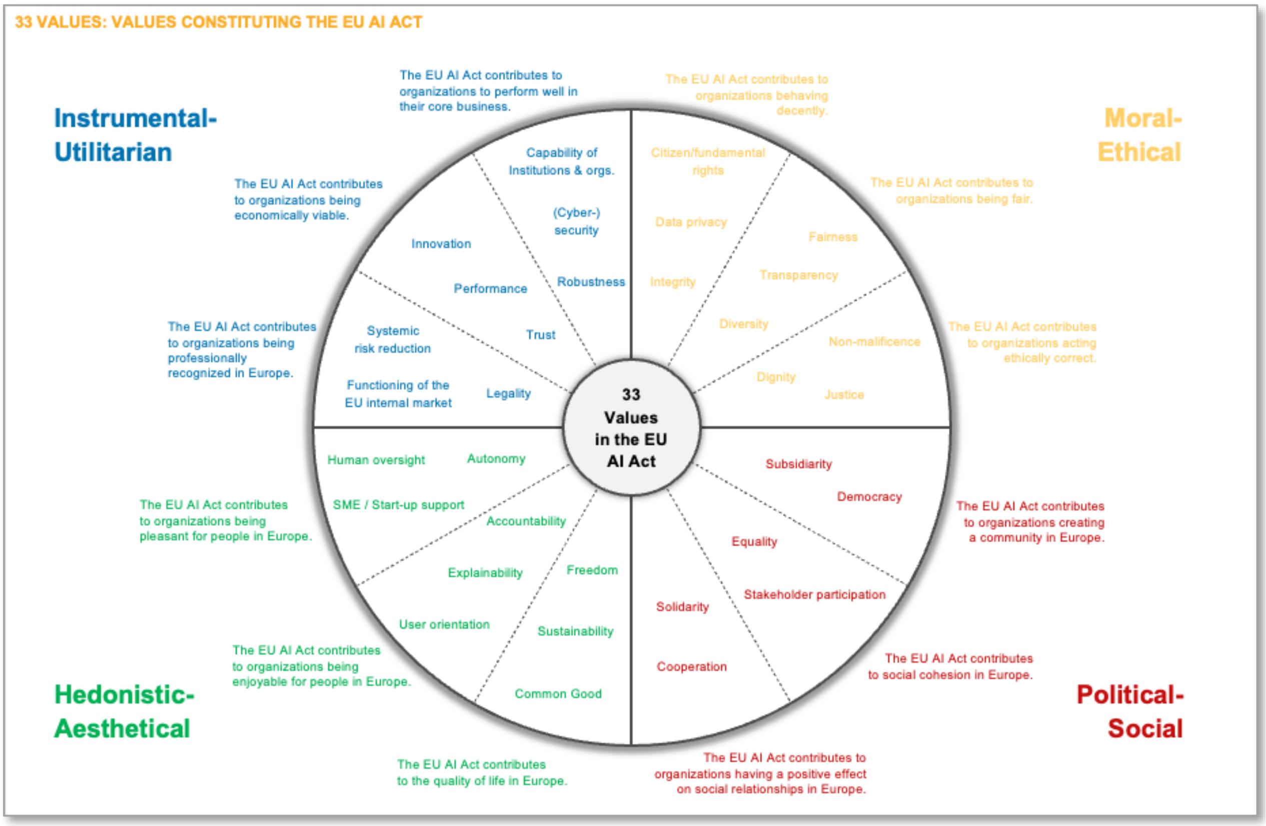


Figure 2, Public Value AI landscape; Source: Own representation

1. **Instrumental-Utilitarian Dimension:**
    a. The EU AI Act contributes to organizations being professionally recognized in Europe: The AIA aims to establish professional recognition for organizations, ensuring **legality** (e.g., definitions, codes of conduct, consistency) through clear regulations and distinct responsibilities. It fosters the **functioning of the internal market** (e.g., through standardization and harmonization) by promoting fair competition and harmonized standards. Additionally, it addresses **systemic risk reduction** (e.g., protection of critical infrastructure) by implementing safeguards to mitigate potential adverse impacts of AI technologies on society and the economy.
    b. The EU AI Act contributes to organizations being economically viable: The AIA promotes the economic viability of organizations by encouraging **innovation** through initiatives such as real-world testing, competitive leadership, and support for open-source collaboration. It emphasizes **performance** by ensuring effectiveness, efficiency, flexibility, and accuracy in AI systems. Moreover, it builds **trust** by prioritizing human-centric AI approaches, thereby strengthening user confidence and fostering responsible AI development and deployment.
    c. The EU AI Act contributes to organizations to perform well in their core business: The AIA improves organizational performance in their core business by strengthening institutional and organizational **capabilities** through initiatives such as information literacy programs, training, and establishing clear lines of authority and competencies. It prioritizes **cybersecurity** to protect against cyber threats. Furthermore, it ensures **robustness** by addressing concerns related to intellectual property rights, reliability of AI systems, and the quality of data utilized, thus facilitating a conducive environment for sustainable business excellence.

2. **Moral-Ethical Dimension:**
    a. The EU AI Act contributes to organizations behaving decently: The AIA fosters decent behavior among organizations by emphasizing **integrity** in their operations and interactions. It prioritizes **data privacy** through robust data governance, protection mechanisms, and restrictions on biometric identification to safeguard individuals' privacy rights. It also upholds **citizens' fundamental rights** by ensuring the protection of workers' rights and applying risk management practices to mitigate potential negative impacts of AI technologies on society, thereby promoting ethical and responsible behavior.

b. The EU AI Act contributes to organizations being fair: The AIA promotes **fairness** within organizations by enforcing principles of non-discrimination, unbiasedness, and the protection of minorities, while prohibiting practices such as social scoring. It emphasizes **transparency** through measures like traceability and the establishment of high-risk AI databases to ensure accountability and trustworthiness. Furthermore, it values **cultural diversity**, recognizing the importance of varied perspectives in developing inclusive and equitable AI systems.

c. The EU AI Act contributes to organizations acting ethically correct: The AIA promotes ethical behavior within organizations by upholding principles of **human dignity**, particularly ethical considerations, as foundational to AI. It emphasizes **non-maleficence** by prioritizing the prevention of harm and adhering to moral principles in AI applications. Additionally, it underscores **justice** through adherence to the rule of law and fostering responsibility among organizations, ensuring that AI technologies are developed and used in ways that align with ethical standards and societal values.

3. Political-Social Dimension:

a. The EU AI Act contributes to social cohesion in Europe: The AIA enhances social cohesion in Europe by promoting **stakeholder participation**, ensuring the independence of research and decision-making processes. It promotes **equality** by striving for fair access to AI technologies and opportunities across diverse communities, thereby strengthening solidarity and inclusivity within European society.

b. The EU AI Act contributes to organizations creating a community in Europe: The AIA helps organizations in building a community in Europe by embracing **subsidiarity**, which empowers local entities to make decisions and take actions that best suit their specific contexts and needs within the broader European framework. It also upholds **democracy** by ensuring participatory decision-making processes and promoting the principles of accountability and representation, thereby fostering a sense of belonging and collaboration across the European community.

c. The EU AI Act contributes to organizations having a positive effect on social relationships in Europe: The AIA promotes organizations' positive impact on social relationships in Europe by promoting **solidarity**, including the protection of minorities and marginalized groups, to create a more inclusive and supportive society. Additionally, it encourages **cooperation** among organizations, fostering collaborative efforts to address societal challenges and promote cohesion across EU communities.

4. **Hedonistic-Aesthetical Dimension:**
   a. <u>The EU AI Act contributes to the quality of life in Europe:</u> The AIA improves the quality of life in Europe by prioritizing **freedom**, ensuring individuals' rights and autonomy in the digital age. It promotes **sustainability** by fostering responsible AI development and deployment practices that mitigate environmental impact and contribute to long-term societal well-being. Moreover, it advances the **common good** by safeguarding public interest, ensuring access to essential private and public services, and promoting social well-being across European communities.
   b. <u>The EU AI Act contributes to organizations being enjoyable for people in Europe:</u> The AIA enhances the enjoyability of organizations for people in Europe by emphasizing **explainability**, ensuring that AI systems are transparent and understandable to users. It promotes **accountability** by holding organizations responsible for their AI applications and their impacts on individuals and society. Additionally, it prioritizes **user-orientation**, designing AI systems with the needs and preferences of users in mind to enhance their overall experience and satisfaction.
   c. <u>The EU AI Act contributes to organizations being pleasant for people in Europe:</u> The AIA aims to make organizations pleasant for people in Europe by supporting **SMEs and startups**, fostering innovation and economic growth while ensuring a diverse marketplace. It emphasizes **human oversight**, preserving human agency and control over AI systems to increase trust and mitigate risks. Additionally, it promotes **autonomy**, empowering individuals, and organizations to make informed decisions and exercise self-determination in their interactions with AI technologies.

Furthermore, we examined the focus of the AIA along the four public value dimensions and the twelve public value items by analyzing code counts. The results show that the AIA places a clear focus on the instrumental-utilitarian dimension. More than half of all the value categories and codes found can be found in the utilitarian-instrumental dimension, with 700 mentions. The other three public value dimensions total 561 mentions. These are distributed as follows: Moral-Ethical (315), Hedonistic-Aesthetic (133) and Political-Social (113). At the item level, a clear focus on the instrumental-utilitarian dimension can also be derived. The three most mentioned value categories all belong to the instrumental-utilitarian dimension, followed by the moral-ethical dimension as shown in the following ranking:

No.	Mentions	PV-D	Public Value Item
1.	377	I-U	The EU AI Act contributes to organizations being professionally recognized in the EU
2.	179	I-U	The EU AI Act contributes to organizations being economically viable
3.	144	I-U	The EU AI Act contributes to organizations to perform well in their core business
4.	135	M-E	The EU AI Act contributes to organizations acting ethically correct
5.	103	M-E	The EU AI Act contributes to organizations being fair
6.	81	H-A	The EU AI Act contributes to organizations being pleasant for people in the EU
7.	77	M-E	The EU AI Act contributes to organizations behaving decently
7.	77	P-S	The EU AI Act contributes to organizations having a positive effect on social relationships in the EU
9.	34	H-A	The EU AI Act contributes to the quality of life in the EU
10.	18	P-S	The EU AI Act contributes to social cohesion in the EU
11.	18	P-S	The EU AI Act contributes to organizations creating a community in the EU
12.	18	H-A	The EU AI Act contributes to organizations being enjoyable for people in the EU

Table 2, Public Value item weights; Source: Own representation based-on Meynhardt & Jasinenko, 2015

To gain insights on which value preferences EU citizens have when regulating AI organizations, we leverage data from the conducted survey. We begin the quantitative analysis by looking at the importance of personal values for the regulation of organizations developing, deploying or using AI. All values were rated considerably closer to the maximum (6) than the minimum (1) of the scale. This implies a high importance of all values included in the sample. The Kolmogorov-Smirnov as well as the Shapiro-Wilk test prove a significant deviation from a normal distribution at the 0.1% significance level. Since the data did not follow a normal distribution, we applied non-parametric tests to analyze value preferences in the following. The Friedman-Test shows that the importance of values for the overall sample varies significantly (Chi-Squared(35)=509.19, $p<.001$, $n=187$). The null hypothesis that the values are equally rated in their importance across the three countries can be rejected at the 0.1% significance level. Pairwise value-comparisons were conducted using Dunn-Bonferroni post-hoc tests. Table 4 shows all value pairs with a statistically significant deviation at the 5% significance level in their valued importance and a strong effect size >0.50 in green as well as with a medium effect >.30 size in yellow (Cohen, 1992).

Value pairs with a small, yet insignificant effect size (r>.10, p>.05) are shown in grey, empty values or not shown value pairs were neither statistically valued significantly different from each other nor had a sufficient effect size. Values on the y-Axis, e.g., (Cyber-)security, Data privacy, Justice, Citizen/fundamental rights etc. were judged to be more important when regulating AI when judged against many of the values on the x-Axis, e.g., Subsidiarity, Capability of institutions and organizations, Stakeholder participation judged to be comparatively less important. Last, we checked for differences between countries in the rated importance of values. For Stakeholder participation (asymptotic Mann-Whitney U-Test r=.90, z=12.367, p=.002, n=187) and Cultural heritage (asymptotic Mann-Whitney U-Test r=.64, z=8.774, p=.012, n=187) countries show significantly differing rated importances with strong effect sizes. A focus on country pairs shows only small effect sizes 0.1 > r < 0.3 but significant deviations. Comparing Finland and Portugal shows that Stakeholder participation (asymptotic Mann-Whitney U-Test r=.28, z=3.248, p=.001, n=133) and Subsidiarity (asymptotic Mann-Whitney U-Test r=.19, z=2.149, p=.032, n=133) are judged to be significantly more important in Finland than Portugal, while for Freedom (asymptotic Mann-Whitney U-Test r=.17, z =1.934, p=.046, n=133) and Sustainability (asymptotic Mann-Whitney U-Test r=.18, z=2.088, p=.037, n=133) the opposite is true. Comparing Germany and Portugal reveals that Stakeholder

Friedman-Test

Ränge

	Mittlerer Rang
Legality	19,57
Functioning of the EU internal market	15,11
Systemic risk reduction	18,01
Innovation	16,37
Performance	17,91
Trust	21,04
Capability of institutions & orgs.	14,18
(Cyber-)security	23,04
Robustness	20,86
Integrity	19,34
Data privacy	22,81
Citizen/fundamental rights	21,92
Fairness	19,74
Transparency	20,38
Diversity	15,88
Dignity	19,10
Non-malificence	21,45
Justice	22,65
Stakeholder participation	14,46
Equality	19,09
Compromise	15,73
Cultural heritage	14,91
Subsidiarity	14,04
Democracy	20,63
Citizen involvement	17,48
Solidarity	18,10
Cooperation	19,05
Freedom	21,02
Sustainability	18,95
Common Good	19,84
Explainability	16,18
Accountability	18,94
User orientation	16,76

Teststatistiken[a]

N	187
Chi-Quadrat	509,194
df	35
Asymp. Sig.	<,001

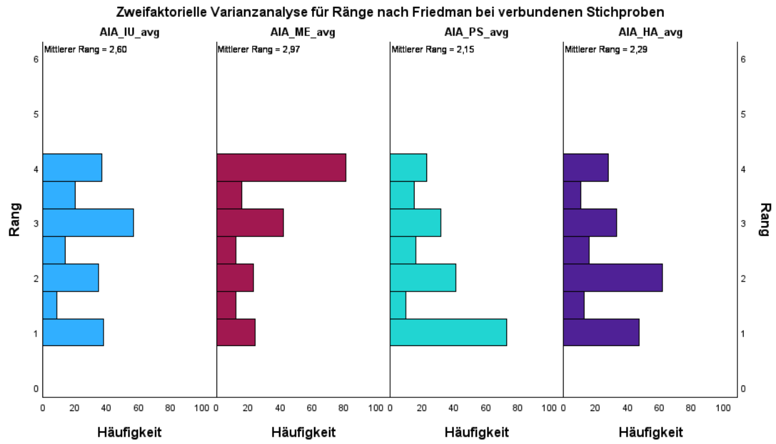
a. Friedman-Test

Table 3 4, Value ranks; Source: Own representation

	Subsidiarity	Capability of institutions &	Stakeholder participation	Cultural heritage	Functioning of the EU	SME / Start-up support	Diversity	Compromise	Autonomy	Explainability	Innovation	User orientation	Citizen involvement
(Cyber-)security	r=0,66 ,p=<.001	r=0,65 ,p=<.001	r=0,63 ,p=<.001	r=0,59 ,p=<.001	r=0,58 ,p=<.001	r=0,58 ,p=<.001	r=0,52 ,p=<.001	r=0,53 ,p=<.001	r=0,51 ,p=<.001	r=0,5 ,p=<.001	r=0,49 ,p=<.001	r=0,46 ,p=<.001	r=0,41 ,p=<.001
Data privacy	r=0,64 ,p=<.001	r=0,63 ,p=<.001	r=0,61 ,p=<.001	r=0,58 ,p=<.001	r=0,56 ,p=<.001	r=0,56 ,p=<.001	r=0,51 ,p=<.001	r=0,52 ,p=<.001	r=0,49 ,p=<.001	r=0,49 ,p=<.001	r=0,47 ,p=<.001	r=0,44 ,p=<.001	r=0,39 ,p=<.001
Justice	r=0,63 ,p=<.001	r=0,62 ,p=<.001	r=0,6 ,p=<.001	r=0,57 ,p=<.001	r=0,55 ,p=<.001	r=0,55 ,p=<.001	r=0,5 ,p=<.001	r=0,51 ,p=<.001	r=0,48 ,p=<.001	r=0,47 ,p=<.001	r=0,46 ,p=<.001	r=0,43 ,p=<.001	r=0,38 ,p=0,001
Citizen/fundament	r=0,58 ,p=<.001	r=0,57 ,p=<.001	r=0,55 ,p=<.001	r=0,51 ,p=<.001	r=0,5 ,p=<.001	r=0,49 ,p=<.001	r=0,44 ,p=<.001	r=0,45 ,p=<.001	r=0,42 ,p=<.001	r=0,42 ,p=<.001	r=0,41 ,p=<.001	r=0,38 ,p=0,001	r=0,32 ,p=0,029
Non-malificence	r=0,54 ,p=<.001	r=0,53 ,p=<.001	r=0,51 ,p=<.001	r=0,48 ,p=<.001	r=0,46 ,p=<.001	r=0,46 ,p=<.001	r=0,41 ,p=<.001	r=0,42 ,p=<.001	r=0,39 ,p=<.001	r=0,39 ,p=<.001	r=0,37 ,p=0,002	r=0,34 ,p=0,011	r=0,29 ,p=0,171
Trust	r=0,51 ,p=<.001	r=0,5 ,p=<.001	r=0,48 ,p=<.001	r=0,45 ,p=<.001	r=0,43 ,p=<.001	r=0,43 ,p=<.001	r=0,38 ,p=0,001	r=0,39 ,p=<.001	r=0,36 ,p=0,004	r=0,36 ,p=0,005	r=0,34 ,p=0,011	r=0,31 ,p=0,054	r=0,26 ,p=0,687
Freedom	r=0,51 ,p=<.001	r=0,5 ,p=<.001	r=0,48 ,p=<.001	r=0,45 ,p=<.001	r=0,43 ,p=<.001	r=0,43 ,p=<.001	r=0,38 ,p=0,001	r=0,39 ,p=<.001	r=0,36 ,p=0,004	r=0,35 ,p=0,006	r=0,34 ,p=0,012	r=0,31 ,p=0,058	r=0,26 ,p=0,736
Robustness	r=0,5 ,p=<.001	r=0,49 ,p=<.001	r=0,47 ,p=<.001	r=0,44 ,p=<.001	r=0,42 ,p=<.001	r=0,42 ,p=<.001	r=0,36 ,p=0,003	r=0,38 ,p=0,002	r=0,35 ,p=0,008	r=0,34 ,p=0,011	r=0,33 ,p=0,023	r=0,3 ,p=0,104	r=0,25 ,p=1
Democracy	r=0,48 ,p=<.001	r=0,47 ,p=<.001	r=0,45 ,p=<.001	r=0,42 ,p=<.001	r=0,4 ,p=<.001	r=0,4 ,p=<.001	r=0,35 ,p=0,008	r=0,36 ,p=0,004	r=0,33 ,p=0,021	r=0,33 ,p=0,027	r=0,31 ,p=0,058	r=0,28 ,p=0,24	r=0,23 ,p=1
Transparency	r=0,46 ,p=<.001	r=0,45 ,p=<.001	r=0,43 ,p=<.001	r=0,4 ,p=<.001	r=0,39 ,p=<.001	r=0,38 ,p=<.001	r=0,33 ,p=0,022	r=0,34 ,p=0,012	r=0,31 ,p=0,057	r=0,31 ,p=0,071	r=0,29 ,p=0,145	r=0,26 ,p=0,557	r=0,21 ,p=1
Human oversight	r=0,45 ,p=<.001	r=0,44 ,p=<.001	r=0,42 ,p=<.001	r=0,39 ,p=<.001	r=0,37 ,p=0,002	r=0,37 ,p=0,002	r=0,32 ,p=0,044	r=0,33 ,p=0,024	r=0,3 ,p=0,107	r=0,3 ,p=0,134	r=0,28 ,p=0,265	r=0,25 ,p=0,967	r=0,2 ,p=1
Fairness	r=0,42 ,p=<.001	r=0,41 ,p=<.001	r=0,39 ,p=<.001	r=0,35 ,p=0,006	r=0,34 ,p=0,014	r=0,34 ,p=0,016	r=0,28 ,p=0,249	r=0,29 ,p=0,146	r=0,26 ,p=0,557	r=0,26 ,p=0,681	r=0,25 ,p=1	r=0,22 ,p=1	r=0,17 ,p=1
Legality	r=0,4 ,p=<.001	r=0,39 ,p=<.001	r=0,37 ,p=0,002	r=0,34 ,p=0,012	r=0,33 ,p=0,027	r=0,32 ,p=0,031	r=0,27 ,p=0,438	r=0,28 ,p=0,263	r=0,25 ,p=0,95	r=0,25 ,p=1	r=0,23 ,p=1	r=0,21 ,p=1	r=0,15 ,p=1
Common good	r=0,42 ,p=<.001	r=0,41 ,p=<.001	r=0,39 ,p=<.001	r=0,36 ,p=0,004	r=0,35 ,p=0,009	r=0,34 ,p=0,011	r=0,29 ,p=0,176	r=0,3 ,p=0,102	r=0,27 ,p=0,4	r=0,27 ,p=0,492	r=0,25 ,p=0,919	r=0,23 ,p=1	r=0,17 ,p=1
Integrity	r=0,39 ,p=<.001	r=0,38 ,p=0,001	r=0,36 ,p=0,005	r=0,32 ,p=0,03	r=0,31 ,p=0,065	r=0,31 ,p=0,075	r=0,25 ,p=0,927	r=0,26 ,p=0,572	r=0,24 ,p=1	r=0,23 ,p=1	r=0,22 ,p=1	r=0,19 ,p=1	r=0,14 ,p=1
Dignity	r=0,37 ,p=0,002	r=0,36 ,p=0,004	r=0,34 ,p=0,013	r=0,31 ,p=0,077	r=0,29 ,p=0,16	r=0,29 ,p=0,184	r=0,24 ,p=1	r=0,25 ,p=1	r=0,22 ,p=1	r=0,21 ,p=1	r=0,2 ,p=1	r=0,17 ,p=1	r=0,12 ,p=1
Equality	r=0,37 ,p=0,002	r=0,36 ,p=0,004	r=0,34 ,p=0,014	r=0,31 ,p=0,081	r=0,29 ,p=0,166	r=0,29 ,p=0,191	r=0,23 ,p=1	r=0,25 ,p=1	r=0,22 ,p=1	r=0,21 ,p=1	r=0,2 ,p=1	r=0,17 ,p=1	r=0,12 ,p=1
Cooperation	r=0,37 ,p=0,003	r=0,36 ,p=0,005	r=0,34 ,p=0,016	r=0,3 ,p=0,092	r=0,29 ,p=0,188	r=0,29 ,p=0,216	r=0,23 ,p=1	r=0,24 ,p=1	r=0,21 ,p=1	r=0,21 ,p=1	r=0,2 ,p=1	r=0,17 ,p=1	r=0,11 ,p=1
Sustainability	r=0,36 ,p=0,004	r=0,35 ,p=0,007	r=0,33 ,p=0,023	r=0,3 ,p=0,13	r=0,28 ,p=0,263	r=0,28 ,p=0,302	r=0,23 ,p=1	r=0,24 ,p=1	r=0,21 ,p=1	r=0,2 ,p=1	r=0,19 ,p=1	r=0,16 ,p=1	r=0,11 ,p=1
Accountability	r=0,36 ,p=0,004	r=0,35 ,p=0,008	r=0,33 ,p=0,024	r=0,29 ,p=0,137	r=0,28 ,p=0,275	r=0,28 ,p=0,316	r=0,22 ,p=1	r=0,24 ,p=1	r=0,21 ,p=1	r=0,2 ,p=1	r=0,19 ,p=1	r=0,16 ,p=1	r=0,11 ,p=1

Table 43: Dunn-Bonferroni test for value pair ranks; Source: Own representation

participation (asymptotic Mann-Whitney U-Test r=.22, z=2.698, p=.007, n=153) and Subsidiarity (asymptotic Mann-Whitney U-Test r=.25, z=3.100, p=.002, n=153) are judged to be significantly more important in Germany than Portugal, and vice versa for Dignity (asymptotic Mann-Whitney U-Test r=.18, z=-.170, p=.030, n=153). For Germany and Finland only the rated importance of Cultural heritage (asymptotic Mann-Whitney U-Test r=.27, z=3.064, p=.002, n=128) shows a significantly higher rated importance in Germany.



Kruskal-Wallis-Test

Ränge

	Country	N	Mittlerer Rang
AIA_IU_avg	DE	76	104,30
	FI	55	106,92
	PT	80	106,99
	Gesamt	211	
AIA_ME_avg	DE	76	100,03
	FI	55	98,10
	PT	80	117,10
	Gesamt	211	
AIA_PS_avg	DE	75	111,07
	FI	55	105,21
	PT	80	100,48
	Gesamt	210	
AIA_HA_avg	DE	76	105,24
	FI	55	98,45
	PT	80	111,91
	Gesamt	211	

Teststatistiken[a,b]

	AIA_IU_avg	AIA_ME_avg	AIA_PS_avg	AIA_HA_avg
Kruskal-Wallis-H	,093	4,306	1,182	1,608
df	2	2	2	2
Asymp. Sig.	,955	,116	,554	,447

a. Kruskal-Wallis-Test
b. Gruppenvariable: Country

Table 5, Non-parametric tests of PV dimension importance; Source: Own representation

On top of the analysis on the item level, the conducted qualitative research enabled a comparison of citizen preferences regarding the Public Value dimensions and sub-dimensions. We calculated the statistical means of items per dimension according to the allocation of items to dimensions and value-items from the qualitative analysis. Reliability tests show a good reliability of the dimensions (Cronbach's α> 0.88 for all dimensions). Comparing the mean ranks applying the Friedman test shows significant differences for the dimensions in their rated importance for AI organization regulation (Chi-Squared(11)=54.646, p<.001, n=210). Pairwise dimension-comparisons with the Dunn-Bonferroni test reveal a significantly higher importance of the moral-ethical dimension compared to all other dimensions (IU asymptotic Dunn-Bonferroni-Test r=0.03, z=0.312, p=.018, n=210; PS asymptotic Dunn-Bonferroni-Test r=0.06, z=0.826, p=.000, n=210; HA asymptotic Dunn-Bonferroni-Test r=0.05, z=0.686, p=.000, n=210). Furthermore, the instrumental-utilitarian was evaluated to be significantly more important than the political-social dimension (asymptotic Dunn-Bonferroni-Test r=0.03, z=0.452, p=.002, n=210). There are no country differences regarding dimension preferences. All effect sizes are not high enough (r<0.1; Cohen, 1992), therefore a relevant difference cannot be implied.

High sub-dimension reliability (Cronbach's $\alpha > 0.8$) was found for all sub-dimensions of the moral-ethical dimension, all other sub-dimensions had a Cronbach's $\alpha > 0.7$. At the subdimension-level the Friedman-Test shows significant importance differences (Chi-Squared(11)=168.188, p<.001, n=208). "The EU AI Act contributes to organizations creating a Community" is significantly least important, while "The EU AI Act contributes to organizations behaving decently" and "The EU AI Act contributes to organizations acting ethically correct" are significantly most important sub-dimensions as shown by pairwise value sub-dimension comparisons.

	AIA_COMMU_avg	AIA_SOCCOH_avg	AIA_PLEASANT_avg	AIA_ENJOY_avg	AIA_PROFESSIONAL_avg	AIA_SOCREL_avg
AIA_DECENT_avg	r=0,19 ,p=<.001	r=0,19 ,p=<.001	r=0,16 ,p=<.001	r=0,15 ,p=<.001	r=0,12 ,p=<.001	r=0,12 ,p=<.001
AIA_ETHICAL_avg	r=0,19 ,p=<.001	r=0,18 ,p=<.001	r=0,16 ,p=<.001	r=0,15 ,p=<.001	r=0,12 ,p=<.001	r=0,11 ,p=<.001
AIA_QOL_avg	r=0,13 ,p=<.001	r=0,13 ,p=<.001	r=0,11 ,p=<.001			
AIA_PERFORM_av	r=0,12 ,p=<.001	r=0,12 ,p=<.001	r=0,1 ,p=0,004			

Table 6, Dunn-Bonferroni test of PV sub-dimension pairwise comparison; Source: Own representation

Table 6 shows all value pairs with a small effect size >0.10 and statistical significance at the 5% level in yellow. There were no significant importance differences with a strong or medium effect size. Values on the x-Axis, e.g., Decent, Ethical, Quality of Life, Performance were judged to be more important when regulating AI when judged against many of the value-items on the y-Axis, e.g., Community, Social cohesion, Pleasant, Enjoyable, Professional, Social Relationships judged to be comparatively less important. When comparing countries on the judged impact along the Public Value sub-dimensions we find Germany valuing "The EU AI Act contributes to organizations creating a Community" higher than Finland (asymptotic Mann-Whitney U-Test z=1.996, p=.046, n=130) and Portugal (z=2.375, p=.018, n=155), yet with a small effect size of r=.18 and r=.19. Furthermore, "The EU AI Act contributes to Organizations behaving decently" and "The EU AI Act contributes to organizations acting ethically correct" is significantly higher valued in Portugal compared to Germany (Decent: r=.17, z=2.080, p=.038, n=155; Ethical: r=.20, z=2.436, p=.015, n=156) and Finland (Decent: r=.19, z=2.246, p=.025, n=134; Ethical: r=.18, z=2.036, p=.042, n=135). Last, "The EU AI Act contributes to the quality of life in Europe" is more important for the Portuguese compared to the Finnish sample (r=.27, z=3.126, p=.002, n=134).

In addition to value differences, we analyzed the relation of the sample towards AI and its regulation. The mean score for the relation towards AI is 3.79 (SD=0.97) and the mean score for regulation of AI is 3.93 (SD=0.87). Both variables show a tendency towards a positive evaluation. The Kruskal-Wallis and Mann-Whitney non-parametric test for distribution across countries shows that there are no significant differences between the countries in their relation to AI (p=.239) and AI regulation (p=.546).

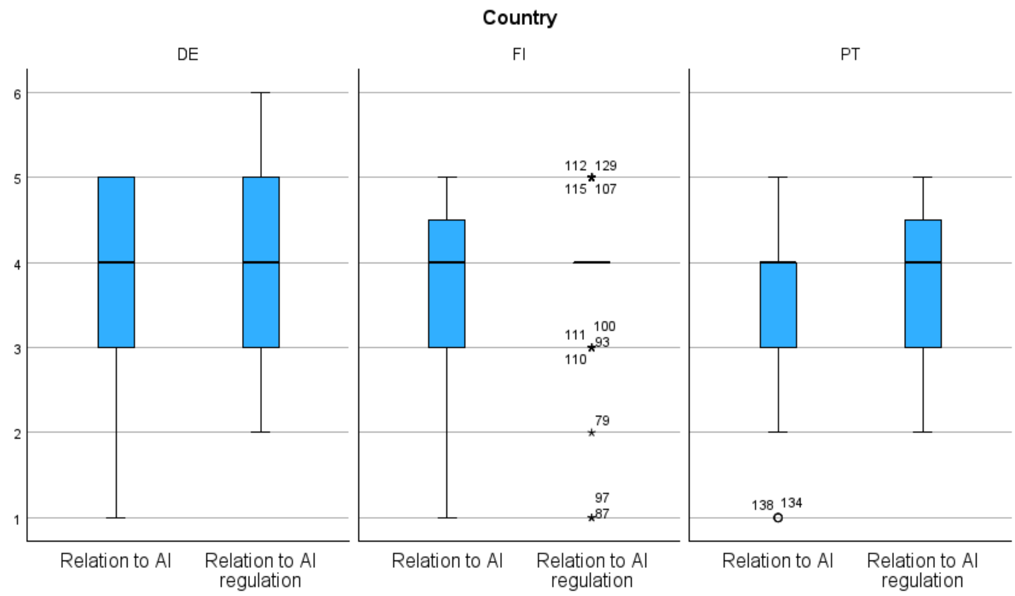


Figure 3, Relation to unregulated and regulated AI across countries; Source: Own representation

Besides the relation to AI and regulated AI the behavioral intention to use, purchase and recommend the unregulated and regulated technology was tested (n=204). Here we focus on the intended use, which showed the highest difference. The intent to use regulated AI is more positive, with a higher mean around 4.10 (MD=4.00, SD=0.891) vs. 3.05 (MD=3.00, SD=1.384) This reflects a more favorable view towards the use of regulated AI technologies. The intended use of regulated AI is significantly higher than of unregulated AI demonstrated by an asymptotic Wilcoxon-Test (z= 8.407, p<.001, n=207) with an effect size of r=.58 which is a strong effect according to Cohen (1992). This strong effect can be found also within Portugal (asymptotic Wilcoxon-Test r=.69, z= 6.120, p<.001, n=79) and Germany (asymptotic Wilcoxon-Test r=.59, z=5.093, p<.001, n=73), while Finland shows a medium effect size (asymptotic Wilcoxon-Test r=.36, z= 2.653, p=.008, n=55). Subsequent post-hoc tests show that the intended use of regulated vs. unregulated AI is significantly stronger for the Portuguese than the Finnish sample (asymptotic Mann-Whitney-Test z=3,75, p<.001, n=134) at the Bonferroni-corrected significance level of 1.67% and a medium effect size of r=.32 and for the German vs. Finnish sample (asymptotic Mann-Whitney-Test z=1.98, p=.047, n=128) at the uncorrected 5% level with a weak effect size of r=.18. Looking for country-differences regarding the intention to use of regulated and unregulated AI separately shows significant differences for the intended use of unregulated AI between all countries (asymptotic Kruskal-Wallis-Test H=23.015, p<.001, n=208) where Finland has the highest mean rank of 134.01 followed by Germany 104.15 and Portugal 84.53. Mann-Whitney post-hoc tests show significant differences between all country pairs. Finland vs. Portugal has a medium effect size of r=.41 (asymptotic Mann-Whitney-Test z=4.85, p<.001, n=135), while Finland vs. Germany (asymptotic Mann-Whitney-Test z=2.79, p=.005, n=128) and Germany vs. Portugal (asymptotic Mann-Whitney-Test z=2.01, p=.045, n=153) have a rather weak effect sizes of r=.24 and r=.16 respectively (Cohen, 1992).

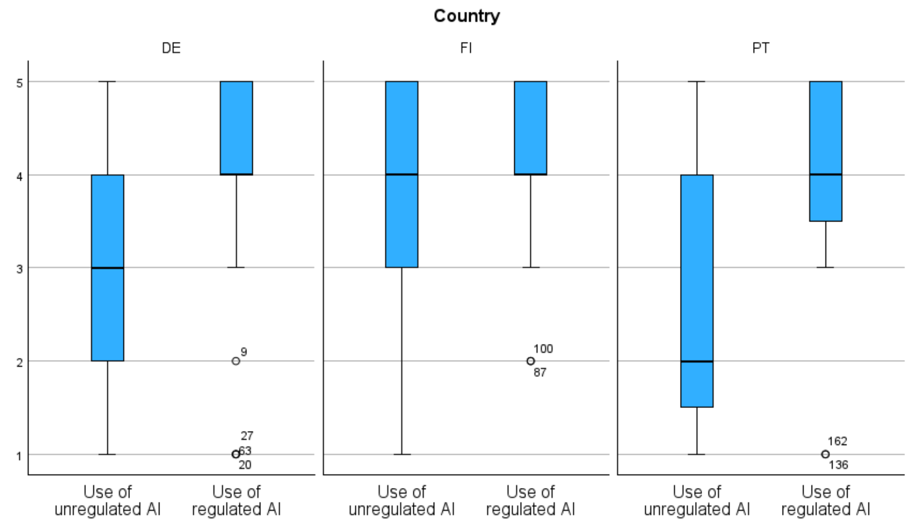


Figure 4, Behavioral intention to use unregulated and regulated AI across countries; Source: Own representation

5 Discussion and conclusion

The EU is the first global actor to adopt a hard law approach towards AI. The AIA tabled by the European Commission in 2021 and adopted by the Council and Parliament in 2024 establishes harmonized rules for AI development, deployment and use across all EU countries, business sectors and types of AI systems. Its primary objective is to enhance the common good by protecting EU citizens from adverse effects of the technology without hampering innovation. Ethical guidelines for Trustworthy AI developed by the HLEG (2019), that can be viewed as part of a proliferation of principles by virtually all stakeholders of AI aiming to influence AI development with their normative guidance, served as the basis for the AI Act. By this, the EU counteracts the criticism of AI ethics' ineffectiveness resulting from its non-binding nature, missing governance tools and infiltration by the private sector. Research conducted by academia and the EU has informed the regulatory process with multiple studies on preferences of public institutions, private organizations and CSOs/NGOs. However, focus should be on the ultimate arbiter of value: citizens, as there is a paucity of evidence on the specific regulatory preferences of the European public. Identifying which values do currently and should guide the regulation of AI organizations from a societal perspective is imperative to achieve acceptance of the regulation and technology and thus its adoption. Our analysis has revealed 33 value categories that constitute the AIA and allocated them to the four value dimensions and twelve sub-dimensions. Subsequently these values, extended by key values from AI ethics guidelines and Public Value literature were evaluated by the EU public represented by samples in Germany, Portugal and Finland regarding their importance when regulating AI organizations.

5.1 Practical and theoretical implications

The results of this study demonstrate that PVT can be leveraged to improve regulation of AI technology and complement AI ethics as a foundation for decision-making. The PV approach helps to reorient the focus back to the individual, to understand value as situated in relationships and overcome the need of choosing between varying value preferences of different stakeholders by grounding judgements on evaluated impact against basic needs. Furthermore, the PV Scorecard provides governments and organizations a valid measurement tool to understand the value impact of their regulatory decisions and business choices.

Several preference patterns have been identified among citizens regarding the regulation of AI organizations. A comparison of the value focus of the AIA with the preferences of the Portuguese, German and Finnish sample reveals that the AIA prioritizes the Instrumental-Utilitarian and Moral-Ethical dimensions (figure 5).

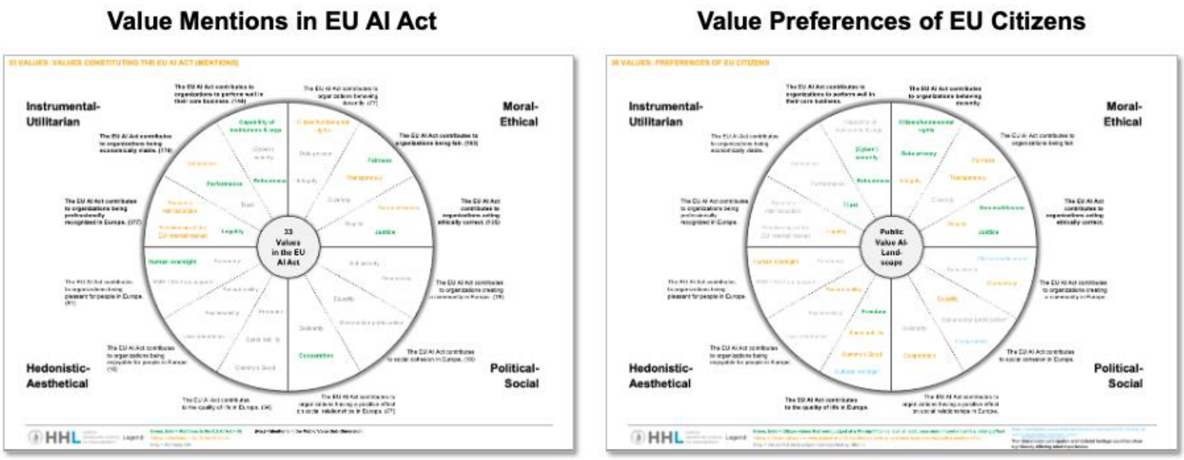


Figure 5, EU AI Act vs. Values Preferences of EU Citizens, Source: Own representation

Based on the mentions of the individual values in the AIA, it can be inferred that the Hedonistic-Aesthetic and Political-Social dimensions are underrepresented. The relevance assessment in our sample indicates that citizens desire a more balanced representation of AI regulation values and that values are judged relevant across all dimensions. In specific, values such as Freedom, Sustainability and Common good in the hedonistic-aesthetic dimension and values such as Citizen involvement, Equality and Democracy are seen as relevant by EU citizens but are rarely addressed in the AIA. In addition, there is an overall need for regulation to promote moral and ethical values. In particular, the need for companies to act ethically and decently in the context of AI appears to be high from a public perspective. In contrast, the AIA places greater emphasis on the instrumental-utilitarian dimension, with particular attention to the professional perception of European organizations. Moreover, we found values of significant importance to European citizens that were not sufficiently addressed in the AIA: Citizen involvement, Compromise and Cultural heritage. This represents a shortcoming.

A comparison of the values desired by different stakeholders revealed that citizens in our sample also expressed a desire for values that were not prioritized by other stakeholders, including governmental organizations, the private sector, and experts. For instance, Fundamental rights (strong preference effect), Sustainability (medium preference effect), and Democratic values (medium preference effect) neglected by private players in comparison to other stakeholders are considered as important by the public. In addition, the AIA features several value categories that are not included in AI ethics guidelines and PVT. These include Trust, Systemic risk reduction, Performance, SME/Start-up support, Functioning of the (internal) market, Stakeholder participation, and Capability of institutions & organizations. In sum, the AIA is guided by an extensive set of values that represents citizen needs in a more holistic way than ethics guidelines. For private players this means that incorporating public values in their product development, deployment and use could increase consumer adoption. The decision by Microsoft (Schiffer & Newton, 2023) and ChatGPT (Samuel, 2024) to lay-off their AI ethics/value alignment teams could therefore potentially lead to unfulfilled societal needs, which would decrease public value and spark societal resistance against their products.

Next, the HLEG has postulated that all promoted values are of equal importance, which could lead to (political) conflicts. Based-on PVT we propose to modify this statement by stating that all basic needs must be met, while different value categories have a differing impact on need fulfillment. The revealed value preferences can guide public and private decision-makers and thus increase the acceptance of their choices. Looking at the value dimensions, citizens assign higher importance to values of the moral-ethical dimension compared to all other dimensions and higher importance to the instrumental-utilitarian versus the political-social dimension. Interestingly, these dimension preferences were found for Portugal, Germany and Finland which implies that there is a common ground in the EU. For sub-dimensions "The EU AI Act contributes to organizations creating a Community" was found to be significantly least important, while "The EU AI Act contributes to organizations behaving decently" and "The EU AI Act contributes to organizations acting ethically correct" are significantly most important. However, country-differences exist. Germany values "The EU AI Act contributes to organizations creating a Community" higher than Finland and Portugal. "The EU AI Act contributes to Organizations behaving decently” and “The EU AI Act contributes to organizations acting ethically correct” is more important in Portugal compared to Germany and Finland. Last, “The EU AI Act contributes to the quality of life in Europe” is more relevant for the Portuguese compared to the Finnish sample. On the item level, Stakeholder participation and Subsidiarity are significantly least important across countries and more important in Finland and Germany than in Portugal. Freedom and Sustainability are more important in Portugal than Finland and Dignity more important than in Germany. Germany values Cultural heritage higher than Finland. When communicating about their regulation governments should try to factor-in these preferences to increase the effectiveness of their efforts to drive policy acceptance. Even though the results concern the regulation of AI organizations we expect that value preferences can give AI companies a good indication which values their products and services should follow.

Finally, the findings of Tallberg and colleagues (2024) from non-state stakeholders preferring regulated compared to unregulated AI cannot be replicated for citizens on the affect-level. However, the intention to use and recommend regulated AI products and services is significantly higher than unregulated AI products and services for the German, Portuguese and Finnish sample. This is despite Finnish people having less doubts to use unregulated AI compared to the other two countries. In its quest to drive adoption of the technology by EU companies and citizens the EU therefore has taken the right path.

5.2 Limitations and outlook

Our study explored values and principles constituting the EU AI Act and tested their relevance for citizens from Portugal, Germany, and Finland. Several shortcomings could lead to a restriction of our results. First, two methodological limitations may potentially compromise the rigorous procedure and the reproducibility of the applied qualitative content analysis. This type of methodology is new to the analysis of legal documents therefore replicability of coding could be limited. Moreover, the selection of categories from PVT for the coding and categorization of the values and principles of the AIA was carried out for the first time. Further research and validation are required to confirm the reliability and validity of this approach. Second, the assignment of the values was based on the knowledge of the researchers and the definitions of the public value dimensions but should be scrutinized or confirmed in subsequent research.

In the second research step, we collected empirical data via survey from three European countries (Portugal, Germany, and Finland) resulting in three potential empirical limitations. First, our study only covers a small number of European citizens and cannot represent the total number of European societies. The sample is not representative in this sense and further research should include other European countries and increase the absolute number of participants in the survey to be able to make representative statements about all European citizens. Second, our study was conducted with a non-probabilistic sample. This could lead to the omission of specific parts of the population. Especially the high share of academics and job definitions as mainly cognitive or analytical raises concern about the representativity of our sample. In consequence, the estimates obtained are potentially not statistically projectable to the population and could compromise the generalizability and validity of your results. Furthermore, the findings could be affected by a self-selection bias towards individuals who have a particular interest in, experience with, or perspective on AI. Moreover, it was necessary to exclude more than 15% of the responses from the sample due to responses being provided in less than five minutes, or incorrect answers for control questions. To validate the findings of our initial sample, it is necessary to distribute the survey across other platforms in the same structure. Last, we identified potential analytical limitations when analyzing the survey data. Given that the data did not follow a normal distribution, non-parametric tests were employed, including the Kruskal-Wallis test and the Mann Whitney test with their respective limitations.

Overall, the findings of our study indicate that Public Value Theory can be applied to legal texts and that the Public Value Scorecard can add value to public value judgements of policies. Moreover, we expect the acceptance of AI regulation and adoption of AI technology to increase with value congruence between citizens and the respective legal texts. Perceived value congruence between humans has been tested to lead to higher trust (Kehoe & Ponting, 2003) and acceptance of modern technologies (Siegrist, 2000) - we recommend testing the same for legal texts and in respect of AI. Furthermore, as previously stated, we recommend further research to be conducted about Public Value Theory and AI regulation. We expect the level of acceptance to increase resulting from a better understanding of what the public values.

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Appendix

Public Value Dimension	Items	Categories	Codes	Count
(Meynhardt 2009, 2012)	(Meynhardt & Jasinenko, 2020)	based on qualitative content analysis of EU AI Act	based on qualitative content analysis of EU AI Act	in EU AI Act
Instrumental-Utilitarian	The EU AI Act contributes to organizations being professionally recognized in Europe.	Legality: Legal certainty, clear responsibility & governance	Risk-based binding rules	31
			Copyright	7
			AI Office	49
			Scientific Panel of Independent Experts	12
			Advisory Forum	7
			EU AI Board	18
			Clarity	12
			Implementation guidelines	6
			Codes of conduct	18
			Compliance	48
			Conformity assessment	33
			Law enforcement	8
			Penalization	5
			Effective remedy	8
			Legal certainty	15
			Migration management	12
			Consistency	16
			Actuality	8
		Functioning of the internal market	Standardization	11
			Harmonized internal market rules	18
			Free movement of goods and services	5
		Systemic risk reduction	General purpose AI	24
			Protection of critical infrastructure	6
	The EU AI Act contributes to organizations being economically viable.	Innovation	Real-world testing / Sandboxes	17
			Open-source	8
			Exceptions to prohibited AI practices	16
		Performance	Timeliness	17
			Appropriateness	52
			Effectiveness	38
			Cost-effectiveness	10
			Accuracy	9
		Trust		12
	The EU AI Act contributes to organizations to perform well in their core business.	(Cap)ability of institutions and organizations	Information and Literacy	35
			Competence, authority and training	27
			Objectivity	15
		(Cyber-)security		16
		Robustness	Reliability	12
			Access to relevant, high-quality data	12
			Training, validation, testing of data	14
			Quality management system	6
			Post-market monitoring system	7
Moral-Ethical	The EU AI Act contributes to organizations behaving decently.	Integrity		5
		Data privacy	Data governance	5
			Data protection	5
			Restriction of "real-time" biometric identification	14
		Citizen / fundamental rights	Application of risk-management	19
			Fundamental rights impact assessment	6
			Protection of worker's rights	9
			Protection of children rights incl. education	9
			Consumer protection	5
	The EU AI Act contributes to organizations being fair.	Fairness	Non-discrimination	34
			Prohibition of untargeted facial / emotion recognition	5
			Non-Profiling (actual behavior, not predicted behavior)	6
			Restricition on "Real-Time" biometric identification	14
			Unbiasedness	8
			Protection of minorities	6
		Transparency	Traceability	19
			High-risk AI databases	6
		Gender & cultural diversity		5
	The EU AI Act contributes to organizations acting ethically correct.	Human dignity	HLEG ethical principles	6
		Non-malificence	Prevention of harm	41
			Protection of Health and Safety	6
		Justice	Rule of law	41
			Responsibility	41
Political-Social	The EU AI Act contributes to social cohesion in Europe.	Stakeholder participation	Independence of research	6
		Equality		12
	The EU AI Act contributes to organizations creating a community in Europe.	Subsidiarity		6
		Democracy		12
	The EU AI Act contributes to organizations having a positive effect on social relationships in Europe.	Solidarity	Protection of minorities	6
		Cooperation		71
Hedonistic-Aesthetical	The EU AI Act contributes to the quality of life in Europe.	Freedom		9
		Sustainability		15
		Common good	Public interest	10
	The EU AI Act contributes to organizations being enjoyable for people in Europe.	Explainability		5
		Accountability		7
		User-orientation		6
	The EU AI Act contributes to organizations being pleasant for people in Europe.	SME & Start-up support	Compliance support	8
		Human oversight	Human agency	16
			Control	43
		Autonomy		14

Appendix-Figure I, 33 Values of the EU AI Act – Overview

Public Value Dimension / Category	Items	Categories	Codes	Count	Quotes
(Meynhardt 2009, 2012)	(Meynhardt & Jasinenko, 2020)	based on qualitative content analysis of EU AI Act	based on qualitative content analysis of EU AI Act	based on qualitative content analysis of EU AI Act	based on qualitative content analysis of EU AI Act
Instrumental-Utilitarian	The EU AI Act contributes to organizations being professionally recognized in Europe.	Legality: Legal certainty, clear responsibility & governance	Risk-based binding rules	31	In order to introduce a proportionate and effective set of binding rules for AI systems, a clearly defined risk-based approach should be followed. That approach should tailor the type and content of such rules to the intensity and scope of the risks that AI systems can generate. It is therefore necessary to prohibit certain unacceptable artificial intelligence practices, to lay down requirements for high-risk AI systems and obligations for the relevant operators, and to lay down transparency obligations for certain AI systems." (p. 23) "High-risk AI systems should only be placed on the Union market, put into service or used if they comply with certain mandatory requirements. Those requirements should ensure that high-risk AI systems available in the Union or whose output is otherwise used in the Union do not pose unacceptable risks to important Union public interests as recognised and protected by Union law (…) the general rule is that several pieces of the EU legislation, such as Regulation (EU) 2017/745 on Medical Devices and Regulation (EU) 2017/746 on In Vitro Diagnostic Devices or Directive 2006/42/EC on Machinery, may have to be taken into consideration for one product, since the making available or putting into service can only take place when the product complies with all applicable Union harmonisation legislation." (p. 33)
			Copyright	7	"Any use of copyright protected content requires the authorization of the rightholder concerned unless relevant copyright exceptions and limitations apply." (p. 61) "In order to increase transparency on the data that is used in the pre-training and training of general purpose AI models, including text and data protected by copyright law, it is adequate that providers of such models draw up and make publicly available a sufficiently detailed summary of the content used for training the general purpose model." (p.62)
			AI Office	49	"(…)t he AI Office may take the necessary actions to monitor the effective implementation and compliance with this Regulation by providers of general purpose AI models, including adherence to approved codes of practice." (p. 217) "The AI Office shall encourage and facilitate the drawing up of codes of practice at Union level as an element to contribute to the proper application of this Regulation." (p. 173) "has been tasked with the development of a template for a questionnaire which deployers will be able to use to meet the relevant requirements." (p.5)
			Scientific Panel of Independent Experts	12	"A scientific panel of independent experts will provide technical advice and input to the AI Office and market surveillance authorities. The scientific panel also has a key role in the enforcement of rules for GPAI models, as it will be able to launch qualified alerts of possible risks to the AI Office." (p.7) "The independent experts constituting the scientific panel should be selected on the basis of up-to-date scientific or technical expertise in the field of artificial intelligence and should perform their tasks with impartiality, objectivity and ensure the confidentiality of information and data obtained in carrying out their tasks and activities." (p.82)
			Advisory Forum	7	"An advisory forum will provide stakeholder input to the Commission (including the AI Office) and to the AI Board. It will comprise a balanced selection of stakeholders, including industry, startups, SMEs, civil society and academia ." (p.7-8) "In fulfilling its role as set out in paragraph 1, the advisory forum may prepare opinions, recommendations and written contributions upon request of the Board or the Commission" (p.194)
			EU AI Board	18	"The Board should reflect the various interests of the AI eco-system and be composed of representatives of the Member States. The Board should be responsible for a number of advisory tasks, including issuing opinions, recommendations, advice or contributing to guidance on matters related to the implementation of this Regulation, including on enforcement matters, technical specifications or existing standards regarding the requirements established in this Regulation and providing advice to the Commission and the Member States and their national competent authorities on specific questions related to artificial intelligence." (p.81) "The European Artificial Intelligence Board should support the Commission, to promote AI literacy tools, public awareness and understanding of the benefits, risks, safeguards, rights and obligations in relation to the use of AI systems." (p.21) "The Board shall be composed of one representative per Member State. (…) Member States shall ensure that their representatives in the Board: have the relevant competences and powers in their Member State so as to contribute actively to the achievement of the Board's tasks, are designated as a single contact point vis-à-vis the Board (…), are empowered to facilitate consistency and coordination between national competent authorities in their Member State" (p.189)
			Clarity	12	High-risk systems need to explain their actions, decisions and functioning: "Certificates issued by notified bodies in accordance with Annex VII shall be drawn-up in a language which can be easily understood." (p.158) "To achieve that objective, rules regulating the placing on the market, putting into service and use of certain AI systems should be laid down, thus ensuring the smooth functioning of the internal market and allowing those systems to benefit from the principle of free movement of goods and services. These rules should be clear and robust (…)." (p.14) " (…) the Commission shall specify that standards have to be consistent, clear and aimed at ensuring that AI systems (…) meet the relevant requirements laid down in this regulation." (p.154) "This explanation should be a clear and meaningful and should provide a basis for affected persons to exercise their rights." (p.90) The expected level of performance metrics should be declared in the accompanying instructions of use. Providers are urged to communicate this information to deployers in a clear and easily understandable way, free of misunderstandings or misleading
			Implementation guidelines	6	"The Commission shall, after consulting the AI Board, and no later than 18 months after the entry into force of this Regulation, provide guidelines specifying the practical implementation of this article completed by a comprehensive list of practical examples of high risk and non-high risk use cases on AI systems in accordance with the conditions set out in Article 82a." (p.112) "(…) in particular on: (a) the application of the requirements and obligations referred to in Articles 8 - 15 and Article 28; (b) the prohibited practices referred to in Article 5; (c) the practical implementation of the provisions related to substantial modification; (d) the practical implementation of transparency obligations laid down in Article 52; (e) detailed information on the relationship of this Regulation with the legislation referred to in Annex II of this Regulation as well as other relevant Union law, including as regards consistency in their enforcement; (f) the application of the definition of an AI system as set out in Article 3(1)." (p.237)
			Codes of conduct	18	"Compliance with these requirements can be achieved through codes of practice, which will be developed by the industry, with the participation of the Member States (through the AI Board), and facilitated by the AI Office." (p.6) "To enable enforcement of this Regulation and create a level-playing field for operators, and taking into account the different forms of making available of digital products, it is important to ensure that, under all circumstances, a person established in the Union can provide authorities with all the necessary information on the compliance of an AI system." (p.53) "Providers that place general purpose AI models on the EU market should ensure compliance with the relevant obligations in this Regulation." (p.62)
			Compliance	48	"Providers of such systems will be obliged to provide all the information and elements to downstream providers of high-risk AI systems so that they can comply with the respective requirements, including for the purpose of conformity assessment." (p.7)
			Conformity assessment	33	"In demonstrating the compliance of a high-risk AI system with the requirements set out in Chapter 2 of this Title, the provider shall follow the conformity assessment procedure set out in Annex VII (…)." (p.156 "The Commission shall ensure that, with regard to high-risk AI systems, appropriate coordination and cooperation between notified bodies active in the conformity assessment procedures pursuant to this Regulation are put in place and properly operated in the form of a sectoral group of notified bodies." (p.152) "In order to ensure a high level of trustworthiness of high-risk AI systems, those systems should be subject to a conformity assessment prior to their placing on the market or putting into service." (p.69)
			Law enforcement	8	"The use of those systems (real-time biometric identification) for the purpose of law enforcement should therefore be prohibited, except in exhaustively listed and narrowly defined situations, where the use is strictly necessary to achieve a substantial public interest, the importance of which outweighs the risks." (p.27) " A specific safeguard procedure should be set for ensuring adequate and timely enforcement against AI systems presenting a risk to health, safety and fundamental rights." (p.84) "The compromise agreement preserves all the exceptions for law enforcement authorities that were added in the Council's revised mandate, including derogation from conformity assessment and the possibility to start real world testing of high-risk AI systems without prior authorisation." (p.4)
			Penalization	5	"The amounts of penalties for the infringements of the various aspects of the AI Act have been fine tuned in Article 71." (p.9) "In compliance with the terms and conditions laid down in this Regulation, Member States shall lay down the rules on penalties and other enforcement measures, which may also include warnings and non-monetary measures, applicable to infringements of this Regulation by operators, and shall take all measures necessary to ensure that they are properly and effectively implemented and taking into account the guidelines issued by the Commission pursuant to Article 82b." (p.225) "The supply of incorrect, incomplete or misleading information to notified bodies and national competent authorities in reply to a request shall be subject to administrative fines of up to 7 500 000 EUR or, if the offender is a company, up to 1 % of its total worldwide annual turnover for the preceding financial year, whichever is higher." (p.226)
			Effective remedy	8	" (…)right to an effective remedy and to a fair trial, right of defence and the presumption of innocence, (…)." (p.33)
			Legal certainty	15	"To ensure legal certainty, it is necessary to clarify that, under certain specific conditions, any distributor, importer, deployer or other third-party should be considered a provider of a high-risk AI system and therefore assume all the relevant obligations." (p.53) "The notion of general purpose AI models should be clearly defined and set apart from the notion of AI systems to enable legal certainty." (p.58) "Differing national rules may lead to fragmentation of the internal market and decrease legal certainty for operators that develop, import or use AI systems." (p.12)
			Migration management	12	"The use of AI systems in migration, asylum and border control management should in no circumstances be used by Member States or Union institutions, agencies or bodies as a means to circumvent their international obligations under the Convention of 28 July 1951 relating to the Status of Refugees as amended by the Protocol of 31 January 1967, nor should they be used to in any way infringe on the principle of non-refoulement, or deny safe and effective legal avenues into the territory of the Union, including the right to international protection." (p.42) "In particular, law enforcement, border control, immigration or asylum authorities should be able to use information systems, in accordance with Union or national law, to identify a person who, during an identity check, either refuses to be identified or is unable to state or prove his or her identity, without being required by this Regulation to obtain prior authorisation. " (p.28)
			Consistency	16	"As part of Union harmonisation legislation, rules applicable to the placing on the market, putting into service and use of high-risk AI systems should be laid down consistently with Regulation (EC) No 765/2008 of the European Parliament and of the Council (…)." (p.51)
			Actuality	8	"The risk management system shall be understood as a continuous iterative process planned and run throughout the entire lifecycle of a high-risk AI system, requiring regular systematic review and updating." (p.116) "This threshold should be adjusted over time to reflect technological and industrial changes, such as algorithmic improvements or increased hardware efficiency, and should be supplemented with benchmarks and indicators for model capability." (p.64)
		Functioning of the internal market	Standardization	11	"Standardisation should play a key role to provide technical solutions to providers to ensure compliance with this Regulation, in line with the state of the art, to promote innovation as well as competitiveness and growth in the single market." (p.68) "(…) the Commission has requested, pursuant to Article 10(1) of Regulation 1025/2012, one or more European standardisation organisations to draft a harmonised standard for the requirements set out in Chapter 2 of this Title; (…)." (p.154)
			Harmonized internal market rules	18	"Differing national rules may lead to fragmentation of the internal market and decrease legal certainty for operators that develop, import or use AI systems. A consistent and high level of protection throughout the Union should therefore be ensured in order to achieve trustworthy AI (…)." (p.12) "In order to ensure an appropriate and effective enforcement of the requirements and obligations set out by this Regulation, which is Union harmonisation legislation, the system of market surveillance and compliance of products established by Regulation (EU) 2019/1020 should apply in its entirety. " (p.83)
			Free movement of goods and services	5	"This regulation ensures the free movement of AI-based goods and services cross-border, thus preventing Member States from imposing restrictions on the development, marketing and use of Artificial Intelligence systems (AI systems), unless explicitly authorised by this Regulation." (p.12) "Consistently with the objectives of Union harmonisation legislation to facilitate the free movement of products in the internal market (…)." (p.33)
		Systemic risk reduction	General purpose AI	24	"These new rules introduce horizontal obligations for all GPAI models, which will include keeping up-to-date and making available, upon request, technical documentation to the AI Office and national competent authorities. They also include providing certain information and documentation to downstream providers for the purpose of compliance with the AI Act. There are some additional requirements for models with systemic risks, which would include performing model evaluation, making risk assessments and taking risk mitigation measures, ensuring an adequate level of cybersecurity (…)." (p.5-6)
			Protection of critical infrastructure	6	"It is thus appropriate that under exceptional reasons of public security or protection of life and health of natural persons, environmental protection and the protection of key industrial and infrastructural assets, market surveillance authorities could authorise the placing on the market or putting into service of AI systems which have not undergone a conformity assessment." (p.71)

Appendix-Figure II, Instrumental-Utilitarian Values (I/II) – incl. Quotes from EU AI Act

Public Value Dimension / Category	Items	Categories	Codes	Count	Quotes
(Meynhardt 2009, 2012)	(Meynhardt & Jasinenko, 2020)	based on qualitative content analysis of EU AI Act	based on qualitative content analysis of EU AI Act	based on qualitative content analysis of EU AI Act	based on qualitative content analysis of EU AI Act
Instrumental-Utilitarian	The EU AI Act contributes to organizations being economically viable.	Innovation	Real-world testing / Sandboxes	17	"In order to avoid fragmentation across the Union, the Commission shall adopt an implementing act detailing the modalities for the establishment, development, implementation, operation and supervision of the AI regulatory sandboxes." (p.179)
			Open-source	8	"The providers of general purpose AI models that are released under a free and open source license, and whose parameters, including the weights, the information on the model architecture, and the information on model usage, are made publicly available should be subject to exceptions as regards the transparency-related requirements imposed on general purpose AI models, unless they can be considered to present a systemic risk (…)." (p.61) "The obligations laid down in this Regulation shall not apply to AI systems released under free and open source licences unless they are placed on the market or put into service as high-risk AI systems or an AI system that falls under Title II and IV." (p.96)
			Exceptions to prohibited AI practices	16	"The use of those systems for the purpose of law enforcement should therefore be prohibited, except in exhaustively listed and narrowly defined situations, where the use is strictly necessary to achieve a substantial public interest, the importance of which outweighs the risks. Those situations involve the search for certain victims of crime including missing people; certain threats to the life or physical safety of natural persons or of a terrorist attack; and the localisation or identification of perpetrators or suspects of the criminal offences referred to in Annex IIa if those criminal offences are punishable in the Member State concerned by a custodial sentence or a detention order for a maximum period of at least four years and as they are defined in the law of that Member State." (p.27)
		Performance	Timeliness	17	"In order to accelerate the process of development and placing on the market of high-risk AI systems listed in Annex III, it is important that providers or prospective providers of such systems may also benefit from a specific regime for testing those systems in real world conditions, without participating in an AI regulatory sandbox." (p.77) "Member States shall ensure that competent authorities referred to in paragraphs 1 and 1a allocate sufficient resources to comply with this Article effectively and in a timely manner." (p.175)
			Appropriateness	52	"The measures adopted by the providers to comply with the mandatory requirements of this Regulation should take into account the generally acknowledge state of the art on artificial intelligence, be proportionate and effective to meet the objectives of this Regulation." "It is appropriate that, in order to minimise the burden on operators and avoid any possible duplication, for high-risk AI systems related to products which are covered by existing Union harmonisation legislation following the New Legislative Framework approach, the compliance of those AI systems with the requirements of this Regulation should be assessed as part of the conformity assessment already foreseen under that legislation." (p.70) "Notified bodies shall perform their activities while avoiding unnecessary burdens for providers, and taking due account of the size of an undertaking, the sector in which it operates, its structure and the degree of complexity of the high risk AI system in question." (p.148)
			Effectiveness	38	"Notified bodies shall have sufficient internal competences to be able to effectively evaluate the tasks conducted by external parties on their behalf." (p.146) "On top of that, this Regulation aims to strengthen the effectiveness of such existing rights and remedies by establishing specific requirements and obligations, including in respect of transparency, technical documentation and record-keeping of AI systems." (p.15) "To prevent the circumvention of this Regulation and to ensure an effective protection of natural persons located in the Union, this Regulation should also apply to providers and deployers of AI systems that are established in a third country, to the extent the output produced by those systems is intended to be used in the Union." (p.21)
			Cost-effectiveness	10	"To ensure consistency and avoid unnecessary administrative burden or costs, providers of a product that contains one or more high-risk artificial intelligence system, (…), should have a flexibility on operational decisions on how to ensure compliance of a product that contains one or more artificial intelligence systems with all applicable requirements of the Union harmonised legislation in a best way." (p.33) "The Commission should regularly assess the certification and compliance costs for SMEs including start-ups, through transparent consultations deployers and should work with Member States to lower such costs. For example, translation costs related to mandatory documentation and communication with authorities may constitute a significant cost for providers and other operators, notably those of a smaller scale." (p.79)
			Accuracy	9	"It is therefore appropriate to classify as high-risk, insofar as their use is permitted under relevant Union and national law, a number of AI systems intended to be used in the law enforcement context where accuracy, reliability and transparency is particularly important to avoid adverse impacts (…)." (p.41) "Requirements should apply to high-risk AI systems as regards risk management, the quality and relevance of data sets used, technical documentation and record-keeping, transparency and the provision of information to deployers, human oversight, and robustness, accuracy and cybersecurity. Those requirements are necessary to effectively mitigate the risks for health, safety and fundamental rights (…)." (p.45) "High-risk AI systems should perform consistently throughout their lifecycle and meet an appropriate level of accuracy, robustness and cybersecurity, in the light of their intended purpose and in accordance with the generally acknowledged state of the art." (p.49)
		Trust		12	"The purpose of this Regulation is (…) to promote the uptake of human centric and trustworthy artificial intelligence while ensuring a high level of protection of health, safety, fundamental rights enshrined in the Charter, including democracy and rule of law and environmental protection, against harmful effects of artificial intelligence systems in the Union and to support innovation (…)." (p.12) "Given the major impact that artificial intelligence can have on society and the need to build trust, it is vital for artificial intelligence and its regulatory framework to be developed according to Union values enshrined in Article 2 TEU, the fundamental rights and freedoms enshrined in the Treaties, the Charter. As a pre-requisite, artificial intelligence should be a human-centric technology." (p.13) "(…) it is important to recall the 2019 Ethics Guidelines for Trustworthy AI developed by the independent High-Level Expert Group on AI (HLEG) appointed by the Commission. In those Guidelines the HLEG developed seven non-binding ethical principles for AI which should help ensure that AI is trustworthy and ethically sound." (p.24)
	The EU AI Act contributes to organizations to perform well in their core business.	(Cap)ability of institutions and organizations	Information and Literacy	35	"In order to obtain the greatest benefits from AI systems while protecting fundamental rights, health and safety and to enable democratic control, AI literacy should equip providers, deployers and affected persons with the necessary notions to make informed decisions regarding AI systems." (p.20) "In particular, in order to minimise the risks to implementation resulting from lack of knowledge and expertise in the market as well as to facilitate compliance of providers, notably SMEs, including start-ups, and notified bodies with their obligations under this Regulation, the AI-on demand platform, the European Digital Innovation Hubs and the Testing and Experimentation Facilities established by the Commission and the Member States at national or EU level should contribute to the implementation of this Regulation." (p.80) "With a view to eliminating or reducing risks related to the use of the high-risk AI system, due consideration shall be given to the technical knowledge, experience, education, training to be expected by the deployer and the presumable context in which the system is intended to be used." (p.117)
			Competence, authority and training	27	"In particular, where appropriate, such measures should guarantee that the system is subject to in-built operational constraints that cannot be overridden by the system itself and is responsive to the human operator, and that the natural persons to whom human oversight has been assigned have the necessary competence, training and authority to carry out that role." (p.49) "Furthermore, deployers should ensure that the persons assigned to implement the instructions for use and human oversight as set out in this Regulation have the necessary competence, in particular an adequate level of AI literacy, training and authority to properly fulfil those tasks." (p.56)
			Objectivity	15	"Notifying authorities shall be established, organised and operated in such a way that no conflict of interest arises with conformity assessment bodies and the objectivity and impartiality of their activities are safeguarded." (p.142) "Notified bodies shall be organised and operated so as to safeguard the independence, objectivity and impartiality of their activities." (145) "The experts shall perform their tasks with impartiality, objectivity and ensure the confidentiality of information and data obtained in carrying out their tasks and activities." (p.195)
		(Cyber-)security		16	"Cybersecurity plays a crucial role in ensuring that AI systems are resilient against attempts to alter their use, behaviour, performance or compromise their security properties by malicious third parties exploiting the system's vulnerabilities." (p.50) "Furthermore, providers should ensure an adequate level of cybersecurity protection for the model and its physical infrastructure, if appropriate, along the entire model lifecycle." (p.66)
		Robustness	Reliability	12	"Similarly, in the health sector where the stakes for life and health are particularly high, increasingly sophisticated diagnostics systems and systems supporting human decisions should be reliable and accurate." (p.33) "High quality data and access to high quality data plays a vital role in providing structure and in ensuring the performance of many AI systems, especially when techniques involving the training of models are used, with a view to ensure that the high-risk AI system performs as intended and safely and it does not become a source of discrimination prohibited by Union law." (p.45) "Such techniques and methods should be sufficiently reliable, interoperable, effective and robust as far as this is technically feasible (…)." (p.73)
			Access to relevant, high-quality data	12	"For the development and assessment of high-risk AI systems, certain actors, such as providers, notified bodies and other relevant entities, such as digital innovation hubs, testing experimentation facilities and researchers, should be able to access and use high quality datasets within their respective fields of activities which are related to this Regulation." (p.46) "Without prejudice to paragraph 1 and 1a, to the extent the deployer exercises control over the input data, that deployer shall ensure that input data is relevant and sufficiently representative in view of the intended purpose of the high-risk AI system." (p.138)
			Training, validation, testing of data	14	"Training, validation and testing datasets shall be relevant, sufficiently representative, and to the best extent possible, free of errors and complete in view of the intended purpose." (p.119) "High quality datasets for training, validation and testing require the implementation of appropriate data governance and management practices. Datasets for training, validation and testing, including the labels, should be relevant, sufficiently representative, and to the best extent possible free of errors and complete in view of the intended purpose of the system." (p.45-46)
			Quality management system	6	"Providers of high-risk AI systems shall put a quality management system in place that ensures compliance with this Regulation. That system shall be documented in a systematic and orderly manner in the form of written policies, procedures and instructions (…)." (p.127) "The provider should establish a sound quality management system, ensure the accomplishment of the required conformity assessment procedure, draw up the relevant documentation and establish a robust post-market monitoring system. " (p.52)
			Post-market monitoring system	7	"The provider should establish a sound quality management system, ensure the accomplishment of the required conformity assessment procedure, draw up the relevant documentation and establish a robust post-market monitoring system." (p.52) "The post-market monitoring system shall actively and systematically collect, document and analyse relevant data which may be provided by deployers or which may be collected through other sources on the performance of high-risk AI systems throughout their lifetime, and allow the provider to evaluate the continuous compliance of AI systems with the requirements set out in Title III, Chapter 2. Where relevant, post-market monitoring shall include an analysis of the interaction with other AI systems." (p.200)

Appendix-Figure III, Instrumental-Utilitarian Values (II/II) – incl. Quotes from EU AI Act

Public Value Dimension / Category	Items	Categories	Codes	Count	Quotes
(Meynhardt 2009, 2012)	(Meynhardt & Jasinenko, 2020)	based on qualitative content analysis of EU AI Act	based on qualitative content analysis of EU AI Act	based on qualitative content analysis of EU AI Act	based on qualitative content analysis of EU AI Act
Moral-Ethical	The EU AI Act contributes to organizations behaving decently.	Integrity		5	"They should carry out their tasks and activities in such a manner as to protect, in particular, intellectual property rights, confidential business information and trade secrets, the effective implementation of this Regulation, public and national security interests, the integrity of criminal or administrative proceedings, and the integrity of classified information." (p.89) "Notified bodies shall be capable of carrying out all the tasks falling to them under this Regulation with the highest degree of professional integrity and the requisite competence in the specific field, whether those tasks are carried out by notified bodies themselves or on their behalf and under their responsibility." (p.146)
		Data privacy	Data governance	5	"Training, validation and testing data sets shall be subject to appropriate data governance and management practices appropriate for the intended purpose of the AI system." (p.118) "The right to privacy and to protection of personal data must be guaranteed throughout the entire lifecycle of the AI system. In this regard, the principles of data minimisation and data protection by design and by default, as set out in Union data protection law, are applicable when personal data are processed. Measures taken by providers to ensure compliance with those principles may include not only anonymisation and encryption, but also the use of technology that permits algorithms to be brought to the data and allows training of AI systems without the transmission between parties or copying of the raw or structured data themselves, without prejudice to the requirements on data governance provided for in this Regulation." (p.47)
			Data protection	5	"The right to privacy and to protection of personal data must be guaranteed throughout the entire lifecycle of the AI system. In this regard, the principles of data minimisation and data protection by design and by default, as set out in Union data protection law, are applicable when personal data are processed." (p.47) "This regulation (…) does not affect the obligations of providers and deployers of AI systems in their role as data controllers or processors stemming from national or Union law on the protection of personal data in so far as the design, the development or the use of AI systems involves the processing of personal data." (p.15)
			Restriction of "real-time" biometric identification	14	"The use of AI systems for 'real-time' remote biometric identification of natural persons in publicly accessible spaces for the purpose of law enforcement is particularly intrusive to the rights and freedoms of the concerned persons, to the extent that it may affect the private life of a large part of the population, evoke a feeling of constant surveillance and indirectly dissuade the exercise of the freedom of assembly and other fundamental rights." (p.27) The following artificial intelligence practices shall be prohibited: (…) the placing on the market or putting into service for this specific purpose, or use of biometric categorisation systems that categorise individually natural persons based on their biometric data to deduce or infer their race, political opinions, trade union membership, religious or philosophical beliefs, sex life or sexual orientation (…) ." (p.106)
		Citizen / fundamental rights	Application of risk-management	19	"The risk management system should consist of a continuous, iterative process that is planned and run throughout the entire lifecycle of a high-risk AI system. This process should be aimed at identifying and mitigating the relevant risks of artificial intelligence systems on health, safety and fundamental rights. The risk management system should be regularly reviewed and updated to ensure its continuing effectiveness, as well as justification and documentation of any significant decisions and actions taken subject to this Regulation." (p.44)
			Fundamental rights impact assessment	6	"The use of the 'real-time' remote biometric identification system in publicly accessible spaces shall only be authorised if the law enforcement authority has completed a fundamental rights impact assessment as provided for in Article 29a and has registered the system in the database according to Article 51." (p.108)
			Protection of worker's rights	9	"Those rights include the right to human dignity, respect for private and family life, protection of personal data, freedom of expression and information, freedom of assembly and of association, and non-discrimination, right to education consumer protection, workers' rights, rights of persons with disabilities, gender equality, intellectual property rights, right to an effective remedy and to a fair trial, right of defence and the presumption of innocence, right to good administration." (p.33 - 34)
			Protection of children rights incl. education	9	"Aside from the many beneficial uses of artificial intelligence, that technology can also be misused and provide novel and powerful tools for manipulative, exploitative and social control practices. Such practices are particularly harmful and abusive and should be prohibited because they contradict Union values of respect for human dignity, freedom, equality, democracy and the rule of law and Union fundamental rights, including the right to non-discrimination, data protection and privacy and the rights of the child." (p.25) "When improperly designed and used, such systems can be particularly intrusive and may violate the right to education (…)." (p.39)
			Consumer protection	5	"Those rights include the right to human dignity, respect for private and family life, protection of personal data, freedom of expression and information, freedom of assembly and of association, and non-discrimination, right to education consumer protection, workers' rights, rights of persons with disabilities, gender equality, intellectual property rights, right to an effective remedy and to a fair trial, right of defence and the presumption of innocence, right to good administration." (p.33 - 34)
	The EU AI Act contributes to organizations being fair.	Fairness	Non-discrimination	34	"The accuracy, non-discriminatory nature and transparency of the AI systems used in those contexts are therefore particularly important to guarantee the respect of the fundamental rights of the affected persons, notably their rights to free movement, non-discrimination, protection of private life and personal data, international protection and good administration." (p.41 - 42)
			Prohibition of untargeted facial / emotion recognition	5	"The placing on the market, putting into service for this specific purpose, or use of AI systems that create or expand facial recognition databases through the untargeted scraping of facial images from the internet or CCTV footage should be prohibited (…)." (p.32) "The notion of emotion recognition system for the purpose of this regulation should be defined as an AI system for the purpose of identifying or inferring emotions or intentions of natural persons on the basis of their biometric data. This refers to emotions or intentions such as happiness, sadness, anger, surprise, disgust, embarrassment, excitement, shame, contempt, satisfaction and amusement." (p.19)
			Non-Profiling (actual behavior, not predicted behavior)	6	"In line with the presumption of innocence, natural persons in the EU should always be judged on their actual behaviour. Natural persons should never be judged on AI-predicted behaviour based solely on their profiling, personality traits or characteristics, such as nationality, place of birth, place of residence, number of children, debt, their type of car, without a reasonable suspicion of that person being involved in a criminal activity based on objective verifiable facts and without human assessment thereof." (p.31)
			Restricition on "Real-Time" biometric identification	14	"This means that the list includes the prohibition of real-time biometric identification by law enforcement authorities in publicly accessible spaces, but with some notable and clearly defined exceptions, as listed in Article 5(1)(d), which are now subject to a range of safeguards, including monitoring and oversight measures and limited reporting obligations at EU level." (p.3)
			Unbiasedness	8	"Results provided by AI systems could be influenced by such inherent biases that are inclined to gradually increase and thereby perpetuate and amplify existing discrimination, in particular for persons belonging to certain vulnerable groups including racial or ethnic groups." (p.46) "In order to protect the right of others from the discrimination that might result from the bias in AI systems, the providers should, exceptionally, to the extent that it is strictly necessary for the purposes of ensuring bias detection and correction in relation to the highrisk AI systems, subject to appropriate safeguards for the fundamental rights and freedoms of natural persons (…)." (p.47)
			Protection of minorities	6	"In addition, AI systems may also otherwise exploit vulnerabilities of a person or a specific group of persons due to their age, disability within the meaning of Directive (EU) 2019/882, or a specific social or economic situation that is likely to make those persons more vulnerable to exploitation such as persons living in extreme poverty, ethnic or religious minorities." (p.25)
		Transparency	Traceability	19	"In order to ensure a level of traceability of the AI system's functioning that is appropriate to the intended purpose of the system, logging capabilities shall enable the recording of events relevant for: (…)." (p.121)
			High-risk AI databases	6	"The Commission shall, in collaboration with the Member States, set up and maintain a EU database containing information referred to in paragraphs 2 and 2a concerning high-risk AI systems referred to in Article 6(2) which are registered in accordance with Articles 51 and 54a." (p.199)
		Gender & cultural diversity		5	"Diversity, non-discrimination and fairness means that AI systems are developed and used in a way that includes diverse actors and promotes equal access, gender equality and cultural diversity, while avoiding discriminatory impacts and unfair biases that are prohibited by Union or national law." (p.24)
	The EU AI Act contributes to organizations acting ethically correct.	Human dignity	HLEG ethical principles	6	"Aside from the many beneficial uses of artificial intelligence, that technology can also be misused and provide novel and powerful tools for manipulative, exploitative and social control practices. Such practices are particularly harmful and abusive and should be prohibited because they contradict Union values of respect for human dignity, freedom, equality, democracy and the rule of law and Union fundamental rights, including the right to non-discrimination, data protection and privacy and the rights of the child." (p.25)
		Non-malificence	Prevention of harm	41	"The assessment should also include the identification of specific risks of harm likely to impact the fundamental rights of these persons or groups." (p.58)
			Protection of Health and Safety	6	"A Union legal framework laying down harmonised rules on artificial intelligence is therefore needed to foster the development, use and uptake of artificial intelligence in the internal market that at the same time meets a high level of protection of public interests, such as health and safety and the protection of fundamental rights, including democracy, rule of law and environmental protection as recognised and protected by Union law." (p.14)
		Justice	Rule of law	41	"Certain AI systems intended for the administration of justice and democratic processes should be classified as high-risk, considering their potentially significant impact on democracy, rule of law, individual freedoms as well as the right to an effective remedy and to a fair trial." (p.42)
			Responsibility	41	"It is appropriate that a specific natural or legal person, defined as the provider, takes the responsibility for the placing on the market or putting into service of a high-risk AI system, regardless of whether that natural or legal person is the person who designed or developed the system." (p.52)
Political-Social	The EU AI Act contributes to social cohesion in Europe.	Stakeholder participation	Independence of research	6	"A balanced representation of interests involving all relevant stakeholders in the development of standards, in particular SME's, consumer organisations and environmental and social stakeholders in accordance with Article 5 and 6 of Regulation 1025/2012 should therefore be encouraged." (p.68)
		Equality		12	"Aside from the many beneficial uses of artificial intelligence, that technology can also be misused and provide novel and powerful tools for manipulative, exploitative and social control practices. Such practices are particularly harmful and abusive and should be prohibited because they contradict Union values of respect for human dignity, freedom, equality, democracy and the rule of law and Union fundamental rights, including the right to non-discrimination, data protection and privacy and the rights of the child." (p.25)
	The EU AI Act contributes to organizations creating a community in Europe.	Subsidiarity		6	"Since the objective of this Regulation cannot be sufficiently achieved by the Member States and can rather, by reason of the scale or effects of the action, be better achieved at Union level, the Union may adopt measures in accordance with the principle of subsidiarity as set out in Article 5 TEU." (p.92)
		Democracy		12	"The purpose of this Regulation is to improve the functioning of the internal market and promoting the uptake of human centric and trustworthy artificial intelligence, while ensuring a high level of protection of health, safety, fundamental rights enshrined in the Charter, including democracy, rule of law and environmental protection against harmful effects of artificial intelligence systems in the Union and supporting innovation." (p.93)
	The EU AI Act contributes to organizations having a positive effect on social relationships in Europe.	Solidarity	Protection of minorities	6	"When assessing for the purposes of paragraph 1 whether an AI system poses a risk of harm to the health and safety or a risk of adverse impact on fundamental rights that is equivalent to or greater than the risk of harm posed by the high-risk AI systems already referred to in Annex III, the Commission shall take into account the following criteria: (…)f) the extent to which there is an imbalance of power, or the potentially harmed or adversely impacted persons are in a vulnerable position in relation to the user of an AI system, in particular due to status, authority, knowledge, economic or social circumstances, or age; (…)." (p.113-114)
		Cooperation		71	"The Commission could develop and recommend voluntary model contractual terms between providers of high-risk AI systems and third parties that supply tools, services, components or processes that are used or integrated in high-risk AI systems, to facilitate the cooperation along the value chain." (p.55)
Hedonistic-Aesthetical	The EU AI Act contributes to the quality of life in Europe.	Freedom		9	"This Regulation should support innovation, respect freedom of science, and should not undermine research and development activity." (p.23)
		Sustainability		15	"The purpose of this Regulation is to improve the functioning of the internal market and promoting the uptake of human centric and trustworthy artificial intelligence, while ensuring a high level of protection of health, safety, fundamental rights enshrined in the Charter, including democracy, rule of law and environmental protection against harmful effects of artificial intelligence systems in the Union and supporting innovation." (p.93)
		Common good	Public interest	10	"This Regulation should provide the legal basis for the providers and prospective providers in the AI regulatory sandbox to use personal data collected for other purposes for developing certain AI systems in the public interest within the AI regulatory sandbox, only under specified conditions (…)." (p.77)
	The EU AI Act contributes to organizations being enjoyable for people in Europe.	Explainability		5	"Transparency means that AI systems are developed and used in a way that allows appropriate traceability and explainability, while making humans aware that they communicate or interact with an AI system, as well as duly informing deployers of the capabilities and limitations of that AI system and affected persons about their rights." (p.24)
		Accountability		7	"The seven principles include: human agency and oversight; technical robustness and safety; privacy and data governance; transparency; diversity, non-discrimination and fairness; societal and environmental well-being and accountability." (p.24)
		User-orientation		6	"The database should also be user-friendly, for example by providing search functionalities, including through keywords, allowing the general public to find relevant information (…)." (p.72)
	The EU AI Act contributes to organizations being pleasant for people in Europe.	SME & Start-up support	Compliance support	8	"Regulatory sandboxes should be widely available throughout the Union, and particular attention should be given to their accessibility for SMEs, including startups." (p.76)
		Human oversight	Human agency	16	"The seven principles include: human agency and oversight; technical robustness and safety; privacy and data governance; transparency; diversity, non-discrimination and fairness; societal and environmental well-being and accountability." (p.24)
			Control	43	"According to the Guidelines of HLEG, human agency and oversight means that AI systems are developed and used as a tool that serves people, respects human dignity and personal autonomy, and that is functioning in a way that can be appropriately controlled and overseen by humans." (p.24)
		Autonomy		14	"AI-enabled manipulative techniques can be used to persuade persons to engage in unwanted behaviours, or to deceive them by nudging them into decisions in a way that subverts and impairs their autonomy, decision-making and free choices." (p.25)

Appendix-Figure IV, Moral-Ethical, Political-Social and Hedonistic-Aesthetical Values – incl. Quotes from EU AI Act